

Tax Incidence with Endogenous Exit

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The measured passthrough of taxes onto consumer prices can be biased when agents may select out of the observed tax base and their exit is correlated with passthrough. The bias depends on the share of the tax base that exits and the difference between measured and counterfactual exiter passthrough. When the latter is unknown, I show how additional information can tighten bounds on passthrough for the original tax base and sign the bias. Applied to the 2018–19 trade war, the framework yields worst-case bounds of 0.897–1, implying a modest correction to the literature’s full-passthrough benchmark.

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1 Introduction

Canonically, the passthrough of a tax onto consumer prices—and hence its incidence on consumers—is governed by the relative elasticities of supply and demand, as well as market structure (Gruber 2019; Weyl and Fabinger 2013). However, two common empirical features make it difficult to assess passthrough following a reform. First, agents typically differ in how much of a tax they pass through. For example, when a sales tax falls on firms with heterogeneous market power, they may systematically differ in how much of the tax they pass on to buyers. Second, agents typically have the option to exit the observed tax base through avoidance, evasion, or shutdown, or other means. If the exiters have systematically different passthrough from those who remain, then the observed tax base reflects survivorship bias, and any resulting passthrough estimates are contaminated by it. This paper develops a sufficient statistics framework for characterizing and bounding selection bias in empirical passthrough estimates.

The framework’s core insight is that correcting selection bias requires accounting for the response of economic activity that is no longer observed in the tax base. Otherwise, measured passthrough—and the buyer-side incidence inferred from it—can rise simply because low-passthrough activity disappears from the data, even when individual passthrough responses remain unchanged. I show that correcting passthrough for this bias requires two quantities: the share of the original tax base that exits, and the difference between survivor passthrough and the average passthrough that would have applied to exiting activity had it remained in the observed tax base. Together with measured survivor passthrough, these two quantities are sufficient to recover the average passthrough of the original tax base—cohort passthrough—without a structural model. For taxes on final consumption goods, this also corrects the consumer incidence attributed to the original tax base. For goods purchased by firms, I show how the incidence interpretation depends on market structure.

However, the average passthrough of the exiter population is rarely known, and I provide a methodology for progressively tightening bounds on the magnitude and sign of the bias in three discrete steps. In the first step, I establish worst-case bounds on cohort passthrough along the lines of Manski (1989), using the share of the original tax base that exits together with a range of possible passthrough. These bounds are typically asymmetric around survivor passthrough. If, for example, the market cannot have a passthrough above one, then passthrough near the upper end of that range leaves more room for overstating than understating buyer-side incidence. Second, the researcher can use information about multiple margins of exit to tighten the bounds. If she knows how much activity exits through each margin and whether its exiters have higher or lower average passthrough than survivors, then the low-passthrough exit share limits how much measured passthrough can overstate cohort passthrough, while the high-passthrough exit

share limits how much it can understate it.¹ However, knowing the size and direction of each exit margin need not establish the sign of the overall bias. Thus, in a third step, I use additional restrictions on exiter passthrough to establish a “dominance” condition under which one margin must outweigh the other. For example, if high-passthrough exit cannot offset the overstatement caused by low-passthrough exit, cohort passthrough must lie below survivor passthrough. The bounds then become one-sided, even though the exact correction remains unknown. Together, these steps also bound buyer-side incidence.

I apply the framework to the 2018–19 trade war between the United States and China. A large literature using bilateral customs data finds passthrough around one for the 2018 U.S. tariffs on China, indicating that the tariffs were largely reflected in the prices paid by U.S. importers (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020; Cavallo et al. 2021). However, standard terms-of-trade logic predicts some absorption through lower prices received by Chinese exporters, and there is some evidence that Chinese exporters left the direct China–U.S. trade sample by shipping to the U.S. via third countries (Iyoha et al. 2025; Freund 2024) while others simply ceased trading with the U.S. Since typical tariff-passthrough estimates condition on activity that remains in direct trade, the framework suggests that selection may help resolve the puzzle if exiting activity would, on average, have passed through less of the tariff than survivors.

Toward characterizing the direction of selection under each exit margin, I develop a simple partial equilibrium trade model with Kimball (1995) demand and variation in marginal cost across exporters. The exporters make a discrete choice about whether to ship directly, transship, or cease serving the U.S. market. The model makes a clear prediction about the direction of selection. Low-cost exporters have larger sales and lower passthrough, and their larger sales make the tariff savings more likely to justify the additional fixed cost of transshipment. High-cost exporters have higher passthrough and are closer to being unable to cover the fixed cost of serving the U.S. market. Consequently, transshippers, which leave the direct China–U.S. trade sample, are selected from the low-passthrough side of the distribution, while the Melitz (2003)-style exit of ceasing trade with the U.S. selects from the high-passthrough side. Thus, as long as we can measure the share of exit due to each margin, it is possible to characterize bounds on cohort passthrough and the sign of bias when combined with restrictions on exiter passthrough.

I measure the two exit margins using harmonized bilateral trade data across countries and more detailed U.S.–China trade flows. To measure transshipment, I build on Freund (2024) by using a series of screens to isolate likely transshipment from China to the United States. This approach complements firm-level evidence from Vietnamese customs records (Iyoha et al. 2025) and yields estimates below several external benchmarks (CEA 2026). I use more granular data to construct a proxy for participation exit, tracking product categories that disappear or contract

1. This usage of monotone selection restrictions to tighten bounds follows Manski and Pepper (2003).

persistently while excluding those associated with substantial detected transshipment. I only allow exit to be detected from 2018–19 and follow it through 2023. By 2023, transshipment accounts for 4.8–8.4 percent of the tariff-exposed sample’s 2017 direct China–U.S. trade value, compared with less than one percent for participation exit. Transshipment is particularly pronounced for business inputs, and extensive robustness and falsification tests support the interpretation that I am detecting tariff-induced transshipment.

I also examine whether the characteristics of exiting activity support the model’s selection mechanism. The model predicts larger sales among transshippers, and smaller sales and higher prices among participation exiters. Products associated with transshipment alone had approximately 39 percent larger pre-tariff sales than products associated with participation exit alone, a statistically significant difference under both screens. The more detailed customs data provide further evidence for the participation mechanism within products: HTS10 categories subsequently flagged for participation exit had smaller sales and quantities and higher pre-tariff unit values than unflagged categories within the same HS6 product and physical unit of measurement. This high-price, low-sales profile is consistent with participation exit coming from higher-cost varieties. Together, these comparisons support the model’s opposing selection predictions and motivate using margin-specific restrictions to tighten the bounds on cohort passthrough.

Finally, I apply the framework’s methodology to bound cohort passthrough, tighten these bounds, and establish the sign of the selection bias. Using the model-implied passthrough range between zero and one, the worst-case bounds, which use the total exit share but no information about its composition, place aggregate cohort passthrough between 0.897 and the survivor benchmark of one. Imposing the direction of selection raises the lower bound slightly to 0.902. At the category level, the worst-case lower bound on capital passthrough is nearly the same as the aggregate, while the lower bound on passthrough for intermediate goods is around 0.85, and for consumption it is 0.97. Next, I put a firmer estimate on the passthrough correction and establish a dominance threshold by using evidence on the passthrough of large firms from Amiti, Itskhoki, and Konings (2019). I calibrate transshippers’ passthrough to 0.525 and participation-exiter passthrough to one, which gives calibrated cohort passthrough of 0.953–0.973, a reduction of approximately 2.7–4.7 percentage points from the survivor estimate. Under competitive domestic production, the corresponding correction to domestic incidence for the original import base is therefore modest in the aggregate, with larger corrections for business inputs. Applying the same calibration implies that aggregate survivor passthrough only has to exceed 0.564 for transshipment exit to dominate participation exit under both screens. This threshold is well below typical estimates for the trade war.

While the calibrated correction is modest in the tariff application, the mechanism is broader. Wherever heterogeneous agents can leave an observed tax base, measured passthrough reflects

the activity that remains and may therefore differ from the average response of the original tax base. The contribution of the paper is to make that selection explicit and show how limited information about exit can bound the bias in measured passthrough and buyer-side incidence.

Related Literature. This paper relates to three distinct strands of literature.

First, I contribute to a large literature on sufficient statistics for the economic consequences of tax reform (Feldstein 1999; Chetty 2009). Whereas Chetty (2009) shows how the resource costs of tax avoidance affect the welfare interpretation of the elasticities of taxable income, I show how selection into exit from the observed tax base changes the interpretation of measured passthrough. This relates closely to Weyl and Fabinger (2013) and Adachi and Fabinger (2022), which establish that the passthrough of taxes onto consumer prices is a sufficient statistic for the incidence of taxes on consumers; my contribution is that this requires a reinterpretation when agents select into exit. This insight is related to Kopczuk et al. (2016), who show that differences in evasion opportunities can make passthrough depend on where a tax is collected.² By adapting methods on partial identification and monotone selection from Manski (1989) and Manski and Pepper (2003), I show how to place progressively tighter bounds on the gap between measured passthrough and incidence-relevant cohort passthrough.

Second, I build on a literature in trade which detects tariff evasion using discrepancies in reported trade and indirect routing (Fisman and Wei 2004; Javorcik and Narciso 2008; Fisman, Moustakerski, and Wei 2008). The closest predecessor is Freund (2024), who identifies product-level trade patterns consistent with transshipment; I adapt her methods to measure this activity alongside participation exit and express both as shares of the original trade base. This approach, which uses publicly available bilateral customs data, is related to other attempts to measure aggregate transshipment as documented in CEA (2026), which similarly use bilateral data. It also complements evidence on tariff-induced transshipment from China through Vietnam from Iyoha et al. (2025) and evidence on physical transshipment through shipping hubs in Do et al. (2025).

Third, I contribute to a growing literature on the empirical effects of the 2018–19 trade war and measurement of passthrough in that period. Many studies find near-complete passthrough of tariffs onto U.S. import prices using bilateral customs data (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020), for which my central calibration implies a modest correction due to selection into exit. Cavallo et al. (2021) study passthrough at both the border and retail level, but only the former estimates are subject to the selection problem identified here since the latter does not condition on continued direct shipments from China. Finally, Alvarez et al. (2023) study

2. The importance of exit opportunities is illustrated by Flaaen, Hortaçsu, and Tintelnot (2020), which finds that country-specific tariffs on washing machines resulted in firms relocating their production and prices not rising as much. Similarly, they show that when tariffs cover nearly all source countries, limiting the gains from relocation, tariff passthrough onto prices is considerably larger.

a sample of firms with continuing buyer–supplier relationships during the trade war and show lower passthrough onto U.S. import prices than in broader observed samples. Their comparison can capture some of the selection studied here, but a relationship can end without the exporter leaving direct trade, and their sample may retain transshippers. Consequently, their findings provide relevant evidence about selection, but the implication for the sign of the survivor–cohort gap remains ambiguous, as discussed in Section 6.

2 Passthrough under Heterogeneity and Exit

The passthrough of a tax into consumer prices generally determines the marginal incidence of the tax onto consumers (Weyl and Fabinger 2013), and under perfect competition, similarly determines incidence on sellers. Because incidence enters the welfare cost of taxes (Adachi and Fabinger 2022), researchers in public finance and trade have devoted considerable effort to estimating passthrough from regressing prices on tax changes. However, three empirically relevant features of markets complicate the interpretation of the resulting coefficient. First, agents within a tax base may have heterogeneous pricing responses. For example, firms may pass through an excise tax differently because of differences in demand conditions, cost structures, or market power. Second, tax changes may induce agents to leave the observed tax base through avoidance, evasion, or ceasing operations. Third, an agent’s propensity to exit may be correlated with its pricing response. In the excise tax example, a beverage producer may be unable to pass through the tax, so the optimal response may be to slightly reformulate the drink so that it is no longer strictly subject to the tax. Consequently, new sales fall outside the observed tax base, resulting in the passthrough of observed survivors differing from the original policy-exposed cohort. The resulting difference between survivor and cohort passthrough is the “selection wedge.”

This section formalizes the selection wedge in three steps. First, I derive it and show how to bound its effect on passthrough when exit is treated as a single aggregate margin. Second, I build on that by decomposing the effects across multiple exit margins, stating a dominance condition for when one type of exit dominates, and showing how margin-specific information tightens the identified set. Finally, I map these results into incidence.

2.1 The Selection Wedge and Aggregate Bounds on Passthrough

This subsection traces measurement bias in the passthrough coefficient from a single market through to aggregate selection. The starting point is a single submarket subject to a small tax change. Under standard assumptions on supply and demand, the tax shifts the equilibrium buyer price according to the passthrough coefficient β , defined as the elasticity of the buyer price with

respect to the gross-of-tax wedge:³

$$\beta \equiv \frac{d \log p^b}{d \log(1 + \tau)}.$$

Under perfect competition, β is determined entirely by the relative magnitudes of the supply and demand elasticities, $\beta = \eta_S / (\eta_S + |\eta_D|) \in [0, 1]$ (Gruber 2019). Under imperfect competition, markups may adjust in equilibrium and thereby affect passthrough. With oligopolistic competition, for example, β depends additionally on the curvature of demand and need not lie below one.

In practice, however, tax instruments rarely apply to a single, well-defined market. Instead, they may be levied over submarkets within a tax base, which may differ in technology, market structure, and other primitives that load onto the curvature of supply and demand. For example, in tariff applications, submarkets may differ in foreign exporter market power, which results in systematic variation across products in how easily exporters can pass the tax onto domestic prices. To summarize these heterogeneous responses, I define aggregate passthrough as a weighted average across the activity originally exposed to the tax. The original tax base matters when we want to know whether firms exposed to a tax generally pass it on to buyers. If firms that would have passed through less instead exit, an estimate based only on continuing transactions can give a misleading answer to that question.

To capture this heterogeneity, consider an original cohort of activity observed before a tax reform, divided into submarkets indexed by $i \in \mathcal{I}$ and distributed according to a measure G . Let ξ_i summarize the technology, market structure, and other conditions that determine submarket i 's local passthrough. Evaluating that response at the post-reform tax rate τ gives a submarket-specific coefficient

$$\beta_i \equiv \beta(\xi_i) = \frac{d \log p_i^b}{d \log(1 + \tau)} \in [\beta_{\min}, \beta_{\max}]. \quad (1)$$

In textbook applications, the interval $[\beta_{\min}, \beta_{\max}]$ would fall between zero and one. I maintain greater generality by allowing $\beta_{\max}$ to exceed one, as is sometimes the case with imperfect competition.

In empirical work, researchers typically cannot directly estimate individual β_i and must instead aggregate across submarkets. The resulting coefficient, which averages across submarkets, depends on which economic activity is represented in the data and how each submarket's price responses are weighted. For concreteness, let ω_i denote submarket i 's normalized share of the

3. For a small ad valorem tax, $d \log(1 + \tau) \approx d\tau$, so this definition is first-order equivalent to the conventional semi-elasticity of the buyer price with respect to the tax rate. A per-unit tax yields the same local object after normalizing the tax change by the pre-tax price.

cohort's pre-reform, tax-exclusive transaction value, with $\int_I \omega_i dG(i) = 1$. I hold these weights fixed when comparing the original cohort with the activity that remains after exit. The selection results below also apply to other fixed, nonnegative weighting schemes. Let

$$\mathbb{E}_\omega[X] \equiv \int_I X(i) \omega_i dG(i)$$

denote the corresponding share-weighted average. Aggregate passthrough for the original cohort is

$$\beta_{\text{cohort}} \equiv \mathbb{E}_\omega[\beta_i] = \int_I \beta_i \omega_i dG(i). \quad (2)$$

Section 2.3 explains when this cohort-level price response maps into incidence and how the relevant weights depend on the tax and its exposure across submarkets.

After the tax reform, however, some activity in the original cohort may have left the observed tax base. Exit may mean literal exit via shutdown, or it may mean increasing avoidance or evasion in response to a higher tax rate. If the same submarket characteristics ξ_i that determine passthrough also affect retention, the resulting passthrough coefficient among survivors, β_{survivor} , may not equal β_{cohort} .⁴

To formalize this problem, and for expositional simplicity, I associate each submarket i with a single policy-exposed agent whose choice determines how much of the activity represented by that submarket remains in the observed tax base. I refer to this agent as firm i , although it could instead be a worker, household, or other market participant. Let $V_i(\sigma; \xi_i, \tau)$ denote firm i 's private payoff under the post-reform tax rate τ from retaining a share $\sigma \in [0, 1]$ of its pre-reform activity, measured by transaction value, in the observed tax base, after optimizing its other choices conditional on σ , including how to allocate the remaining activity among its available alternatives. This payoff incorporates the associated private benefits and costs. For a firm, it may represent profits; for a household, it may represent utility. The agent chooses its retained share to maximize this payoff:

$$\sigma_i(\tau) \in \arg \max_{\sigma \in [0,1]} V_i(\sigma; \xi_i, \tau). \quad (3)$$

The results below do not require a particular specification of these private payoffs; they describe how the resulting retention shares change the composition of observed passthrough. For the remainder of the analysis, I suppress the dependence of the retained share on τ and write it as σ_i . If $\sigma_i = 1$, all of the activity represented by submarket i remains in the observed tax base; if $\sigma_i = 0$, none of it remains. If firms completely exit the observed tax base, that does not necessarily entail shutting down. Avoidance or evasion can instead shift economic activity out of the observed tax

4. Passthrough is homogeneous within each submarket. The comparison holds market conditions fixed rather than solving for the equilibrium that would obtain if all original activity remained in the tax base.

base without the firm shutting down. An interior value permits partial exit, like shifting activity into the gray market or otherwise engaging in partial avoidance or evasion behavior. The total share of pre-reform activity that exits is

$$\theta \equiv 1 - \mathbb{E}_\omega[\sigma_i].$$

Firm choices about how much market activity to retain in the observed tax base alter the composition of activity in the observed data. Holding the pre-reform weights fixed, the retained transaction value associated with submarket i is proportional to $\omega_i \sigma_i$. Provided that $\theta < 1$, the transaction-value-weighted passthrough among retained activity is therefore

$$\beta_{\text{survivor}} \equiv \frac{\mathbb{E}_\omega[\beta_i \sigma_i]}{\mathbb{E}_\omega[\sigma_i]} = \frac{\mathbb{E}_\omega[\beta_i \sigma_i]}{1 - \theta}, \quad (4)$$

where the survivor estimand refers to passthrough among the retained activity from the original tax base. The difference in survivor and cohort passthrough comes down to weighting. I make this comparison holding submarket passthrough fixed, so exiting activity enters the cohort average with the response it would have shown had it remained in the observed tax base. Whereas the cohort estimand uses ω_i , the survivor estimand uses weights proportional to $\omega_i \sigma_i$.⁵ This difference generates a selection wedge. Toward formalizing that, let $\text{Cov}_\omega(\beta_i, \sigma_i)$ denote the covariance between passthrough and retention under the pre-reform transaction-value weights.

Proposition 1 (Selection wedge). *Suppose $0 \leq \theta < 1$. The survivor and cohort passthrough estimands differ by the share-weighted covariance of submarket-level passthrough with retention:*

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \frac{\text{Cov}_\omega(\beta_i, \sigma_i)}{\mathbb{E}_\omega[\sigma_i]} = \frac{\text{Cov}_\omega(\beta_i, \sigma_i)}{1 - \theta}. \quad (5)$$

Proof: Appendix [A.1.1](#).

If passthrough is the same across all submarkets, the covariance is zero, so exit alone creates no gap between survivor and cohort passthrough. The sign of the survivor–cohort wedge is given by $\text{Cov}_\omega(\beta_i, \sigma_i)$, where a positive covariance means that retained activity has higher passthrough than the original cohort and a negative covariance reverses the comparison. When the sign of the covariance is known, either through external evidence or by imposing additional structure on the problem, it is straightforward to bound the corrected passthrough. To characterize this

5. Tax changes could also induce more innocuous reweighting through changes in the composition of transactions. That would generate a separate composition term, which I abstract from here.

bound, for $\theta > 0$ define exit-weighted average passthrough as

$$\beta_{\text{exiter}} \equiv \frac{\mathbb{E}_{\omega}[(1 - \sigma_i)\beta_i]}{\theta}.$$

Because σ_i and $1 - \sigma_i$ partition each submarket's activity, cohort passthrough decomposes as

$$\beta_{\text{cohort}} = (1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\text{exiter}}. \quad (6)$$

When $\theta = 0$, β_{exiter} need not be defined; the exit term is zero, and survivor and cohort passthrough coincide. Since β_{exiter} is an exit-weighted average of the submarket-level coefficients β_i , it inherits the maintained support $[\beta_{\min}, \beta_{\max}]$. Combining this support restriction with the ordering implied by equation (5) yields the following bound.

Proposition 2 (Sign-restricted bound). *Suppose $0 \leq \theta < 1$ and $\beta_i \in [\beta_{\min}, \beta_{\max}]$ for all i . Then*

$$\begin{aligned} \text{Cov}_{\omega}(\beta_i, \sigma_i) \geq 0 &\implies \beta_{\text{cohort}} \in [(1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\min}, \beta_{\text{survivor}}], \\ \text{Cov}_{\omega}(\beta_i, \sigma_i) \leq 0 &\implies \beta_{\text{cohort}} \in [\beta_{\text{survivor}}, (1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\max}]. \end{aligned}$$

Proof: Appendix A.1.2.

For example, suppose $\beta_{\text{survivor}} = 1$ under the benchmark support $[\beta_{\min}, \beta_{\max}] = [0, 1]$. With a nonnegative covariance, β_{cohort} must fall between $1 - \theta$ and 1. Appendix A.1.7 shows that, in the special case of binary monotone-threshold selection, this interval can be further tightened using information about how passthrough varies with the exit share, $\partial\beta_{\text{survivor}}/\partial\theta$, holding submarket passthrough and baseline weights fixed as the selection threshold varies.

However, there may be some circumstances in which it is not possible to sign the covariance. In that case, equation (6) yields a two-sided bound using only the support restriction.

Proposition 3 (Support-only bound). *Suppose $0 \leq \theta < 1$ and $\beta_i \in [\beta_{\min}, \beta_{\max}]$ for all i . Then*

$$\beta_{\text{cohort}} \in [(1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\min}, (1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\max}]. \quad (7)$$

Proof: Appendix A.1.3.

The result follows because $\beta_{\text{exiter}} \in [\beta_{\min}, \beta_{\max}]$. The interval in equation (7) has width $\theta(\beta_{\max} - \beta_{\min})$, so the exit share limits the maximum survivor-cohort discrepancy without determining its direction. Conditional on β_{survivor} , θ , and the maintained support restriction, this is the bounded-outcome identified set when no additional information about exiter passthrough

is available (Manski 1989). Importantly, this identified set also features systematic asymmetry. Since its width is $\theta(\beta_{\max} - \beta_{\min})$, the midpoint of the set is

$$(1 - \theta)\beta_{\text{survivor}} + \theta \frac{\beta_{\min} + \beta_{\max}}{2}. \quad (8)$$

Consequently, accounting for exit would shift the midpoint of the identified set a fraction θ of the way from survivor passthrough toward the midpoint of its support. If β_{survivor} exceeds the support's midpoint, then the identified set creates greater scope for measured passthrough to overstate cohort passthrough. The reverse holds if β_{survivor} is less than the support's midpoint. To see why this matters, consider $\beta_{\text{survivor}} = 0.9$ under the canonical $[0, 1]$ passthrough support. With $\theta = 0.1$, the identified set is given by $\beta_{\text{cohort}} \in [0.81, 0.91]$, so measured passthrough may overstate passthrough by as much as 0.09, but can understate by at most 0.01. However, β_{survivor} remains in the set. The following subsection shows how to leverage economically distinct exit margins to potentially sign the wedge and tighten the identified set.

2.2 Selection with Multiple Exit Margins

The previous subsection analyzes the implications for passthrough when there is only one exit margin—or, equivalently, when all exit is pooled into one aggregate measure. In practice, firms often have the option to exit in a variety of ways. For example, they can shut down entirely, or they can choose to avoid taxes. These alternatives may select activity from different parts of the passthrough distribution, so their contributions to the survivor–cohort gap can offset one another. In this section, I show how information about these distinct exit margins can tighten the bounds on passthrough and establish conditions under which one margin dominates the other.

Suppose now that exit can occur through K distinct margins. Together, these margins account for all original-cohort activity absent from the survivor average, without counting the same activity more than once. Let $\rho_i^{(k)} \in [0, 1]$ denote the share of activity in submarket i that exits through margin k . Using the same original-cohort weights as in the previous subsection, define each margin's exit share and, for margins with $\theta_k > 0$, its average passthrough as

$$\sum_{k=1}^K \rho_i^{(k)} = 1 - \sigma_i, \quad \theta_k \equiv \mathbb{E}_\omega[\rho_i^{(k)}], \quad \bar{\beta}_k \equiv \frac{\mathbb{E}_\omega[\beta_i \rho_i^{(k)}]}{\theta_k}.$$

By construction, $\sum_{k=1}^K \theta_k = \theta$. It is useful to partition the margins into two distinct groups depending on their passthrough relative to β_{survivor} . Collect margins satisfying $\bar{\beta}_k \leq \beta_{\text{survivor}}$ into group L and those satisfying $\bar{\beta}_k > \beta_{\text{survivor}}$ into group H . For each group with a positive exit

share, define

$$\theta_g \equiv \sum_{k \in g} \theta_k, \quad \bar{\beta}_g \equiv \frac{\sum_{k \in g} \theta_k \bar{\beta}_k}{\theta_g}, \quad g \in \{L, H\}.$$

Since $\theta_L + \theta_H = \theta$, Equation (6) implies

$$\begin{aligned} \beta_{\text{survivor}} - \beta_{\text{cohort}} &= \sum_{k=1}^K \theta_k (\beta_{\text{survivor}} - \bar{\beta}_k) \\ &= \theta_L (\beta_{\text{survivor}} - \bar{\beta}_L) - \theta_H (\bar{\beta}_H - \beta_{\text{survivor}}). \end{aligned} \tag{9}$$

The two terms on the final line decompose the survivor–cohort wedge into the contribution from low-passthrough exit and the contribution from high-passthrough exit. This decomposition suggests a general dominance condition: if one exit margin’s passthrough gap multiplied by its exit share is larger than the other’s, then it is straightforward to sign $\text{Cov}_\omega(\beta_i, \sigma_i)$, which gives the sign of the survivor–cohort gap and allows for tighter bounds on passthrough.

Proposition 4 (Full-information dominance bound). *Suppose $\theta_L > 0$ and $\theta_H > 0$. Low-passthrough exit dominates—meaning $\beta_{\text{survivor}} > \beta_{\text{cohort}}$ —if and only if*

$$\beta_{\text{survivor}} > \frac{\theta_L \bar{\beta}_L + \theta_H \bar{\beta}_H}{\theta_L + \theta_H} \iff \frac{\theta_L}{\theta_L + \theta_H} > \frac{\bar{\beta}_H - \beta_{\text{survivor}}}{\bar{\beta}_H - \bar{\beta}_L}. \tag{10}$$

Moreover,

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = (\theta_L + \theta_H) \left[\beta_{\text{survivor}} - \frac{\theta_L \bar{\beta}_L + \theta_H \bar{\beta}_H}{\theta_L + \theta_H} \right]. \tag{11}$$

At equality, survivor and cohort passthrough coincide. Under the reverse inequality, high-passthrough exit dominates and $\beta_{\text{survivor}} < \beta_{\text{cohort}}$.

Proof: Appendix A.1.4.

The proposition establishes a necessary and sufficient condition for signing the survivor–cohort passthrough wedge when exit shares and average counterfactual passthrough by margin are known. However, counterfactual exiter passthrough is rarely observed. Suppose instead that available evidence or maintained assumptions restrict each group’s average passthrough to an interval

$$\bar{\beta}_L \in [\underline{b}_L, \bar{b}_L] \subseteq [\beta_{\min}, \beta_{\text{survivor}}], \quad \bar{\beta}_H \in [\underline{b}_H, \bar{b}_H] \subseteq [\beta_{\text{survivor}}, \beta_{\max}].$$

These intervals may still provide enough information to sign the survivor–cohort gap. Low-passthrough dominance is hardest to establish when $\bar{\beta}_L = \bar{b}_L$, which minimizes the low-passthrough contribution, and $\bar{\beta}_H = \bar{b}_H$, which maximizes the opposing contribution. Symmetrically, high-passthrough dominance is hardest to establish when $\bar{\beta}_L = \underline{b}_L$, which maximizes the opposing

low-passthrough contribution, and $\bar{\beta}_H = \underline{b}_H$, which minimizes the high-passthrough contribution. Define

$$\underline{\beta}_{\text{exiter}} \equiv \frac{\theta_L \underline{b}_L + \theta_H \underline{b}_H}{\theta_L + \theta_H}, \quad \bar{\beta}_{\text{exiter}} \equiv \frac{\theta_L \bar{b}_L + \theta_H \bar{b}_H}{\theta_L + \theta_H}.$$

Given these interval restrictions, it is possible to construct robust dominance bounds for when each type of exit may dominate, which permits researchers to sign the bias.

Corollary 1 (Multi-margin dominance bound with uncertainty). *Suppose $\theta_L > 0$ and $\theta_H > 0$, and the stated interval restrictions hold.*

(i) *If $\beta_{\text{survivor}} > \bar{\beta}_{\text{exiter}}$, then*

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} \geq \theta \left(\beta_{\text{survivor}} - \bar{\beta}_{\text{exiter}} \right) > 0. \quad (12)$$

(ii) *If $\beta_{\text{survivor}} < \underline{\beta}_{\text{exiter}}$, then*

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} \leq \theta \left(\beta_{\text{survivor}} - \underline{\beta}_{\text{exiter}} \right) < 0. \quad (13)$$

Proof: Appendix A.1.5.

The corollary shows that the selection wedge can still be signed—and hence the bounds on passthrough tightened—when the counterfactual exiter passthrough is not point-identified. For either of the two cases, it is sufficient to compare the measured β_{survivor} to the relevant threshold. However, even when neither dominance condition determines the sign of the wedge, the group-specific restrictions remain informative about the level of cohort passthrough. Rearranging Equation (9) gives

$$\beta_{\text{cohort}} = (1 - \theta_L - \theta_H) \beta_{\text{survivor}} + \theta_L \bar{\beta}_L + \theta_H \bar{\beta}_H.$$

Applying the bounds on $\bar{\beta}_L$ and $\bar{\beta}_H$ to this identity gives the following interval.

Proposition 5 (Multi-margin bounds under partial information). *Suppose $0 \leq \theta < 1$ and the stated interval restrictions hold for each group with a positive exit share. Then*

$$\beta_{\text{cohort}} \in \left[(1 - \theta_L - \theta_H) \beta_{\text{survivor}} + \theta_L \underline{b}_L + \theta_H \underline{b}_H, (1 - \theta_L - \theta_H) \beta_{\text{survivor}} + \theta_L \bar{b}_L + \theta_H \bar{b}_H \right]. \quad (14)$$

Moreover, if $\bar{\beta}_L \in [\beta_{\min}, \beta_{\text{survivor}}]$ and $\bar{\beta}_H \in [\beta_{\text{survivor}}, \beta_{\max}]$, then

$$\beta_{\text{cohort}} \in [(1 - \theta_L) \beta_{\text{survivor}} + \theta_L \beta_{\min}, (1 - \theta_H) \beta_{\text{survivor}} + \theta_H \beta_{\max}]. \quad (15)$$

Proof: Appendix [A.1.6](#).

Thus, even without tighter bounds on the group averages, knowing the direction of selection at each margin weakly improves upon the aggregate worst-case bound.

Numerical Example. While the rest of the paper shows how the methodology works in the context of tariffs with multiple margins of exit, it is useful to illustrate the bounding approach with a numerical example. Suppose the economy features passthrough bounded by $[\beta_{\min}, \beta_{\max}] = [0, 1]$, as is true under perfect competition as well as some forms of imperfect competition. Suppose a researcher measures the passthrough of a tax as $\beta_{\text{survivor}} = 0.8$, and suppose that she also measures the exit share of the tax base as $\theta = 0.10$. In that case, she could apply the worst-case bound on passthrough from the previous subsection as

$$\beta_{\text{cohort}} \in [(1 - 0.10)(0.8) + 0.10(0), (1 - 0.10)(0.8) + 0.10(1)] = [0.72, 0.82].$$

Now, suppose the researcher has credibly identified that low-passthrough exit is roughly $\theta_L = 0.04$ of the original cohort, while high-passthrough exit accounts for $\theta_H = 0.06$. This allows her to apply Equation (15) to tighten the bounds to

$$\beta_{\text{cohort}} \in [0.90(0.8) + 0.04(0) + 0.06(0.8), 0.90(0.8) + 0.04(0.8) + 0.06(1)] = [0.768, 0.812],$$

where the researcher uses $\bar{\beta}_L \in [0, 0.8]$ and $\bar{\beta}_H \in [0.8, 1]$. This narrows the bound, but it still contains β_{survivor} and does not sign the wedge.

Finally, suppose additional information implies $\bar{\beta}_L \in [0, 0.2]$, while $\bar{\beta}_H \in [0.8, 1]$. The lower and upper bounds on average passthrough among exiting activity implied by these restrictions are

$$\beta_{\text{-exiter}} = \frac{0.04(0) + 0.06(0.8)}{0.10} = 0.48, \quad \bar{\beta}_{\text{exiter}} = \frac{0.04(0.2) + 0.06(1)}{0.10} = 0.68.$$

Even assigning each group its highest admissible passthrough, average passthrough among exiting activity cannot exceed 0.68. Because $\beta_{\text{survivor}} = 0.8 > \bar{\beta}_{\text{exiter}} = 0.68$, low-passthrough exit must dominate. Equivalently, its contribution to the wedge is at least $0.04(0.8 - 0.2) = 0.024$, while the contribution from high-passthrough exit is at most $0.06(1 - 0.8) = 0.012$. Hence,

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} \geq 0.024 - 0.012 = 0.012 = 0.10(0.8 - 0.68) > 0.$$

Thus, low-passthrough exit dominates even though it accounts for only 40 percent of total exit. Applying Proposition 5 to the same restrictions gives

$$\beta_{\text{cohort}} \in [0.90(0.8) + 0.04(0) + 0.06(0.8), 0.90(0.8) + 0.04(0.2) + 0.06(1)] = [0.768, 0.788].$$

Consequently, information about the composition of exit can materially tighten the worst-case bounds and, as in this example, determine the direction of the selection correction. The next subsection examines the conditions under which these passthrough bounds translate into bounds on tax incidence.

2.3 Mapping Passthrough into Buyer-Side Incidence

Together, Sections 2.1 and 2.2 bound the difference between survivor and cohort passthrough across a rich variety of settings. These bounds have direct implications for tax incidence. If exit is systematically correlated with passthrough, then incidence inferred from measured passthrough may be similarly biased. This is straightforward to observe in some settings by leveraging the envelope theorem result from Weyl and Fabinger (2013) and Adachi and Fabinger (2022) that consumer price passthrough determines the first-order change in consumer surplus, holding other prices and income fixed. This matters when observed passthrough is used to draw conclusions about incidence for the activity initially exposed to the tax, rather than only for the transactions that continue.

To make the connection between passthrough and incidence concrete, consider a tax on a final consumption good. A consumer with income m consumes c_i in each submarket i , using the same submarkets and measure G as in Section 2.1, as well as a numeraire good x . Suppose each submarket faces an ad valorem tax τ , and the after-tax purchase price is $p_i^b = (1 + \tau)p_i$. For simplicity, I assume that the consumer has quasilinear utility over a consumption vector $\mathbf{c}$ and an untaxed numeraire x , i.e., $U(\mathbf{c}, x) = u(\mathbf{c}) + x$, where u is differentiable in all of its arguments. In this setting, if we aggregate across submarkets, then the consumer surplus at an interior allocation is given by

$$CS(\mathbf{p}^b) = \max_{\mathbf{c}} \left\{ u(\mathbf{c}) - \int_I p_i^b c_i dG(i) \right\}.$$

Suppose there is a tax increase. By the envelope theorem, the first-order loss in consumer surplus from higher prices is given by $-dCS = \int_I c_i dp_i^b dG(i)$, where $dp_i^b = p_i \beta_i d\tau$, while the corresponding response of suppliers' net prices is $\frac{d \log p_i}{d \log(1+\tau)} = \beta_i - 1$. It is straightforward to connect this result to tax incidence for consumers. Letting B_i denote the pre-reform tax base and $\omega_i = B_i / \int_I B_j dG(j)$ each submarket's share of that base,⁶ the implied consumer incidence per

6. The exact local consumer surplus calculation uses $B_i = p_i c_i$ at the allocation where passthrough is evaluated. Using historical transaction values instead holds the original tax base fixed. If tax increases differ across submarkets, the corresponding weights are proportional to $B_i d\tau_i$.

dollar of additional tax calculated on the unchanged base is given by

$$\frac{\int_I B_i \beta_i d\tau dG(i)}{\left(\int_I B_i dG(i)\right) d\tau} = \int_I \omega_i \beta_i dG(i) = \beta_{\text{cohort}}. \quad (16)$$

This applies the familiar consumer surplus result from Weyl and Fabinger (2013) and Adachi and Fabinger (2022) to the original tax base. The key distinction is that we typically observe a different quantity, β_{survivor} , and that changes the economic interpretation of incidence. Suppose $\beta_{\text{cohort}} < \beta_{\text{survivor}} \leq 1$. This implies that observed price incidence tends to overstate the burden attributed to consumers in that base and, by extension, understates the incidence that firms would bear through lower net prices.⁷ In this example, the survivor–cohort gap measures how much observed passthrough overstates consumer incidence for the original tax base. Voluntary exit reduces firms’ burden relative to remaining taxed by their net private gains from exit, after accounting for any private costs they may incur.⁸

While the derivation and example concerned a tax on a final consumer good, many types of taxes are not so clean to analyze. As an example, consider a uniform import tariff, with submarkets corresponding to different types of goods. The importer may be either a final consumer, in which case the incidence analysis would mimic the above results, or a firm which transforms imports into final consumption goods. In the latter case, under perfect competition in domestic production, passthrough at the border measures the combined first-order burden on domestic firms and consumers.⁹ Under these conditions, the survivor–cohort gap gives the corresponding correction to incidence for the original import base, although it does not identify how that burden is divided between importers and final consumers. However, when importers charge markups, a tariff-induced contraction in output can generate additional losses, so the combined domestic burden may exceed that implied by border passthrough. In either case, if measured passthrough exceeds cohort passthrough, then using the measured coefficient to calculate incidence for the original import base overstates the contribution of higher import prices to the domestic burden.

In sum, the passthrough correction carries through to incidence following the logic of Weyl and Fabinger (2013) and Adachi and Fabinger (2022), along with attendant concerns about how market structure affects its application on the supply side. The correction, which broadly applies across tax types, distinguishes the incidence inferred from surviving transactions from that asso-

7. The exact difference in incidence depends on market structure. Passthrough is incidence for consumers, but is generally not for firms.

8. This brings up an important distinction along the lines of Chetty (2009). If agents incur private costs to avoid or evade via payments to a third party, then those costs are transfers, and a full social welfare calculation would have to net out those costs.

9. This assumes domestic ownership, holds other input prices fixed, excludes additional domestic tax distortions, and measures private losses before accounting for tariff revenue.

ciated with the original tax base. In particular, measured passthrough can be high because firms that would have passed through less of the tax have exited. However, those firms may still be operating and bear some burden through the private costs of evasion or avoidance. Consequently, the fact that they are no longer in the data does not make it innocuous to ignore them when estimating passthrough and incidence; inferences about either from continuing transactions need not be representative of the original base, and the correction I derive provides bounds on the resulting difference in the price incidence attributed to that original activity.

3 Tariff Passthrough under Endogenous Exit

The remainder of the paper applies the framework of Section 2 to an open puzzle: a large literature finds near-complete passthrough of the 2018 U.S. tariffs on China (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020; Cavallo et al. 2021), despite standard terms-of-trade logic predicting some absorption through lower prices received by Chinese exporters. Because these estimates rely on customs data, they condition on reported direct trade. If tariffs induce exporters to leave that channel, the estimated coefficient is survivor passthrough rather than cohort passthrough.¹⁰ The bounds developed in the previous section require, at minimum, an estimate of how much of the original tax base left direct trade because of the tariffs. I start from the premise that there are two major margins of exit: firms ceasing to trade with the United States due to a standard Melitz (2003) channel, and a transshipment margin through which exporters avoid tariffs by routing trade through a third country (Freund 2024). The model, which is standard aside from the transshipment margin, maps otherwise unobservable passthrough of exiting firms into different locations along the passthrough distribution and therefore helps considerably with the measurement of exit and provides the necessary structure to tighten bounds on passthrough bias.

3.1 Environment

I work in a static partial equilibrium setting with two types of agents: domestic importers and foreign exporters. Foreign exporters compete monopolistically over differentiated varieties indexed by $i \in \mathcal{I}$. Each variety i is analogous to the submarkets in the previous section. Exporters face a common demand function and differ only in marginal cost. Each seller sets a price p_i and reaches U.S. buyers through one of two channels indexed by $r \in \{d, a\}$. If the exporter ships through the direct (d) channel, it faces an ad valorem wedge τ^d . If it chooses to ship indirectly via a third country, it faces a wedge $\tau^a \in [0, \tau^d]$. The gross-of-wedge price under channel r is

10. Firm-level studies such as part of Cavallo et al. (2021) and Flaaen, Hortaçsu, and Tintelnot (2020) avoid origin-based selection by tracking specific goods.

$$\tilde{p}_i = (1 + \tau^r)p_i.$$

Buyers. U.S. importers may purchase varieties for final consumption or as inputs into domestic production. They submit demand for each variety based on its gross-of-wedge price $\tilde{p}_i$ and a price aggregator $\mathbf{P}$,

$$q_i = D(\tilde{p}_i \mid \mathbf{P}).$$

$\mathbf{P}$ is exogenous at the firm level, corresponding to a standard monopolistic competition setup in which each firm is small relative to the market and takes the price distribution of rivals as given. I hold $\mathbf{P}$ fixed throughout the comparative statics. Denoting the demand elasticity as $\varepsilon_i > 1$, I assume that demand is weakly Kimball (1995) with superelasticity

$$\kappa_i(\tilde{p}_i \mid \mathbf{P}) \equiv -\frac{\partial \ln \varepsilon_i(\tilde{p}_i \mid \mathbf{P})}{\partial \ln \tilde{p}_i} \leq 0,$$

which nests CES at $\kappa = 0$ and allows elasticity to rise with price when $\kappa < 0$. The Kimball specification is the standard non-CES generalization in the international pricing literature, motivated by evidence that markups and passthrough vary systematically with firm size and marginal cost in ways constant-markup CES demand cannot generate (Gopinath and Itskhoki 2010; Amiti, Itskhoki, and Konings 2019). The full set of regularity conditions on demand, **D1–D5**, is in Appendix A.2.1.

Sellers. Each foreign exporter $i \in \mathcal{I}$ has constant marginal cost m_i , drawn independently from a continuous distribution G with support $[\underline{m}, \bar{m}] \subset (0, \infty)$ and density $g(m) > 0$. The distribution is independent of policy and of the aggregator $\mathbf{P}$. Since each variety i corresponds to a submarket from Section 2.1, the variation in marginal cost supplies the heterogeneity there summarized by ξ_i .

The exporters face a choice about whether to ship directly to the United States, to ship to the U.S. through a third country (transshipment), or whether to cease trade with the U.S. Each choice is associated with different fixed costs and wedges. If the firm ships directly to the U.S., it must pay a uniform fixed cost $F_d \geq 0$ —standard in heterogeneous firm models since Melitz (2003)—and each purchase faces the gross ad valorem wedge $1 + \tau^d$. The wedge corresponds to U.S. tariffs on imports from China. To partially avoid that, the firm may transship through a third country and instead face a wedge $\tau^a < \tau^d$.¹¹ That requires a fixed cost $F_a \geq 0$, which incorporates the

11. I assign the transshipment decision to the exporter to preserve tractability. Transshipment could also arise from importers searching across origin countries or from trade intermediaries arbitraging tariff differentials. In Supplemental Appendix S.1.1, I show that under efficient bargaining with transferable surplus, the cutoffs and the associated margin-specific selection contributions depend only on the joint surplus across channels, independent of which side bears F_a and F_d .

cost of establishing an alternative channel, including contracting with intermediaries and bearing any legal or reputational risks associated with misdeclared origins. The wedge τ^a includes any statutory tariffs on the third country, but it also may include additional costs proportional to the value of a transshipped good. Supplemental Appendix S.1.3 formally handles additional per-unit transshipment costs, including repackaging or relabeling goods and higher transit costs, while Supplemental Appendix S.1.2 extends the model to heterogeneous fixed costs. As a third option, exporters may choose to not pay any fixed costs and instead cease trade entirely with the United States, which is standard Melitz (2003) exit. I normalize the payoff to not shipping to the U.S. to zero.

To formalize the choice between the three options, let

$$V(m, 1 + \tau^r) \equiv \max_{p \geq m} (p - m)D((1 + \tau^r)p \mid \mathbf{P})$$

denote optimized variable profit in channel r . Given that, exporter i compares payoffs across each of the three options

$$\pi(m_i) = \max \{ V(m_i, 1 + \tau^d) - F_d, V(m_i, 1 + \tau^a) - F_a - F_d, 0 \}, \quad (17)$$

corresponding to direct trade, transshipment, and exit from the U.S. market. Exporters which choose to trade directly with the U.S. correspond to $\sigma_i = 1$ in Section 2.1—meaning that they remain in the original reported direct-trade base—while exporters choosing to transship or cease trade with the U.S. correspond to $\sigma_i = 0$. The next subsection translates buyer demand and exporter supply into choices that lead to, in equilibrium, some trade leaving the observed tax base.

3.2 Pricing, Passthrough, and Selection

An exporter's choice about how and whether to exit depends on payoffs within each of the three options. Payoffs depend on the pricing decision and the degree to which the firm can pass through marginal changes in tariffs. I now characterize the pricing decision, which is simply a standard Lerner condition.

Lemma 1 (Lerner condition). *For each channel $r \in \{d, a\}$ with corresponding wedge τ^r , under D1–D5, the optimal price $p_i^*(m_i; r)$ is interior and satisfies*

$$\frac{p_i^* - m_i}{p_i^*} = \frac{1}{\varepsilon_i(\tilde{p}_i^r \mid \mathbf{P})}, \quad \tilde{p}_i^r = (1 + \tau^r)p_i^*.$$

Moreover: (i) p_i^* strictly increases in m_i ; (ii) p_i^* is weakly decreasing in τ^r (constant under CES); and

(iii) q_i^* strictly decreases in m_i and τ^r .

Proof: Appendix A.2.2.

As is standard in many settings, exporter i sets its markup to be inversely proportional to the demand elasticity at the delivered price. When firms have higher marginal costs, that implies higher prices, lower quantities, and weakly lower markups within each channel. With Kimball demand, the weak inequality becomes strict, and the higher marginal costs correspond to lower markups. That variation across varieties generates variation in the passthrough of tariffs to importer prices. Conditional on operating in channel $r \in \{d, a\}$, the passthrough of tariffs to buyer prices is given by

$$\beta_i \equiv \frac{d \ln \tilde{p}_i}{d \ln(1 + \tau^r)} = \frac{\varepsilon(\tilde{p}_i) - 1}{\varepsilon(\tilde{p}_i) - 1 - \kappa(\tilde{p}_i)}, \quad \tilde{p}_i \equiv (1 + \tau^r)p_i^*(m_i) \quad (18)$$

The following proposition characterizes the range of passthrough and its relationship with marginal cost.

Proposition 6 (Passthrough heterogeneity). *Suppose **D1–D5** hold and **P** is fixed. Conditional on operating in a given channel $r \in \{d, a\}$, passthrough is homogeneous across exporters at $\beta_i = 1$ under CES ($\kappa = 0$). When $\kappa < 0$, $\beta_i \in (0, 1)$ and passthrough is strictly increasing in marginal cost.*

Proof: Appendix A.2.3.

Under standard assumptions, passthrough can vary across varieties, and is bounded between zero and one. However, firms have the option to exit the observed tax base through two separate channels, which means that, following a reform, measured passthrough may differ from cohort passthrough. The reason is that, if we restrict our attention to the observed tax base, then we fail to observe firms which continue to ship to the U.S. via transshipment, as well as firms which cease serving the U.S. market. I now characterize that discrete choice between direct trade, transshipment, and ceasing U.S. sales.

With prices chosen optimally in each channel, exporters make a discrete choice about whether to transship, ship directly, or cease trade with the United States. First consider the choice between shipping directly and transshipping. Denote the gain in variable profit from transshipment as $\Delta V(m) \equiv V(m, 1 + \tau^a) - V(m, 1 + \tau^d)$. Since both channels require a fixed cost F_d , but transshipment requires an additional cost F_a , exporters choose to transship rather than ship directly if and only if $\Delta V(m) > F_a$. Similarly, the exporter compares the profits from shipping in either channel to ceasing trade entirely. Let m_p denote the marginal cost at which direct trade just breaks even; for $m > m_p$, direct trade is unprofitable, but transshipment may still be profitable in that region. Importantly, this can cause direct trade to disappear. To see that, note that at m_p , transshipment profit is $\Delta V(m_p) - F_a$. If this value is positive, then transshipment would be preferable to direct

trade for every exporter with $m < m_p$, since $\Delta V(m)$ decreases with marginal cost, causing no direct trade to exist. To retain direct trade and focus on an interior partition, I impose the following regularity condition:

Assumption 1 (Interior cutoffs). Suppose $\tau^a < \tau^d$ and the fixed costs satisfy

$$\begin{aligned} V(\bar{m}, 1 + \tau^d) &< F_d < V(\underline{m}, 1 + \tau^d), \\ \Delta V(m_p) &< F_a < \Delta V(\underline{m}), \end{aligned}$$

where m_p solves $V(m_p, 1 + \tau^d) = F_d$.

These conditions ensure that both cutoffs lie within the cost distribution and leave a positive mass of direct shippers between them. Given that, we can fully characterize which exporters choose to transship, ship directly, and cease serving the U.S. market.

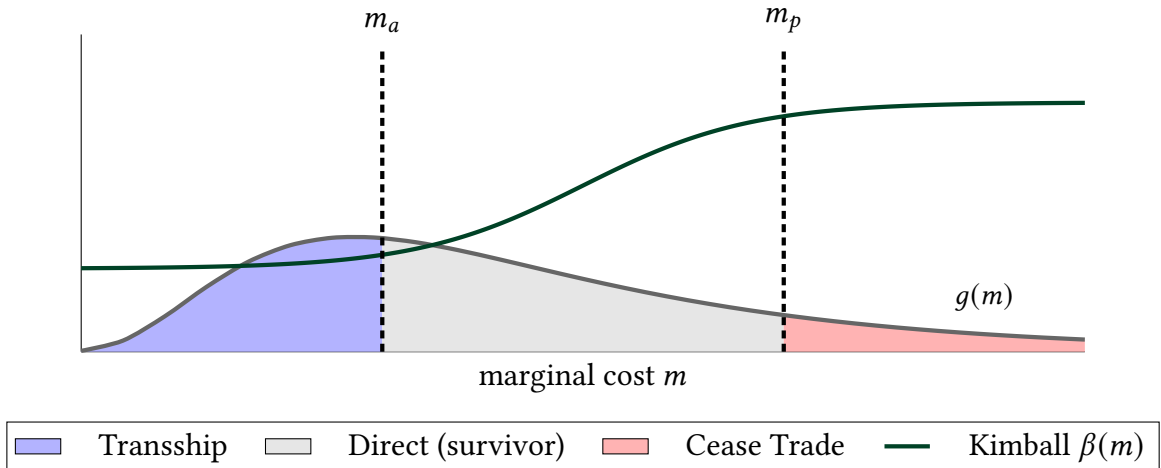
Proposition 7 (Channel partition). Under **D1–D5** and Assumption 1 there exist unique cutoffs $m_a(\tau^d, \tau^a, F_a)$ and $m_p(\tau^d, F_d)$ with $\underline{m} < m_a < m_p < \bar{m}$ such that the cost distribution partitions as

$$\begin{array}{ccc} \underbrace{m < m_a}_{\text{Transship}} & \underbrace{m_a \leq m \leq m_p}_{\text{Direct (survivor)}} & \underbrace{m > m_p}_{\text{Exit the U.S. market}} \end{array}$$

Moreover, tariff increases shrink the survivor set from both sides, i.e., $\frac{\partial m_a}{\partial \tau^d} > 0$ and $\frac{\partial m_p}{\partial \tau^d} < 0$.

Proof: Appendix A.2.4.

Figure 1: Channel Partition under Heterogeneous Passthrough



Note: The distribution $g(m)$ shows the density of marginal costs. The solid green line shows passthrough under Kimball demand. Dashed lines show the cutoffs at the original tariff.

Figure 1 visualizes the partition from the proposition. The blue mass to the left of the cut-off m_a have lower passthrough and choose to transship, whereas varieties with high marginal costs and higher passthrough choose to cease serving the U.S. market to the right of m_p . Under Kimball demand, low-cost exporters have higher markups and earn higher variable profits. Their larger sales volume makes access to the lower tariff wedge relatively more valuable, which allows them to justify paying the additional fixed cost to transship. By Proposition 6, these low-cost exporters also have lower direct-channel passthrough, so the expansion of transshipment selects low-passthrough exporters out of the observed tax base. The opposite is true for the exiters who choose to cease trade with the U.S. When tariffs rise, both cutoffs move toward each other, making it an empirical question which exit margin contributes more to the survivor-cohort gap. To make that more precise, it is useful to map the cutoff movements into the estimands of Section 2. Suppose the tariff τ^d rises, holding fixed the transshipment wedge and maintaining Assumption 1 before and after the increase. Since the pre-increase direct shippers are the “cohort” and the reported direct-trade sample excludes activity that exits that channel, the question is how selection into exit changes measured passthrough.

Here, ω_i denotes exporter i ’s normalized pre-increase share of direct-trade transaction value within the original cohort, and β_i denotes its direct-channel passthrough. For exporters that exit, β_i is the counterfactual passthrough that would obtain if they remained in direct trade. Let $\sigma_i^{(a)}$ indicate whether an original-cohort variety switches from direct trade to transshipment, and $\sigma_i^{(p)}$ whether it ceases serving the U.S. market following an increase in τ^d . Define

$$\theta^a = \mathbb{E}_\omega[\sigma_i^{(a)}], \quad \theta^p = \mathbb{E}_\omega[\sigma_i^{(p)}],$$

With $\sigma_i = 1 - \sigma_i^{(a)} - \sigma_i^{(p)}$ indicating continued direct trade, average passthrough among the original-cohort exporters that remain in the direct channel is

$$\beta_{\text{survivor}} = \frac{\mathbb{E}_\omega[\beta_i \sigma_i]}{\mathbb{E}_\omega[\sigma_i]} = \frac{\mathbb{E}_\omega[\beta_i \sigma_i]}{1 - \theta^a - \theta^p}.$$

β_{survivor} is what researchers typically measure using bilateral trade data, but with exit, β_{cohort} is generally not point-identified from β_{survivor} alone in studies like Amiti, Redding, and Weinstein (2019) and Fajgelbaum et al. (2020). However, we can tighten from worst-case bounds by leveraging the composition of selection. Let $\bar{\beta}_a$ denote the average passthrough of new transshippers and $\bar{\beta}_p$ the corresponding value for varieties which cease trade. When $\kappa < 0$ and both exit shares are positive, the monotone partition implies

$$\bar{\beta}_a < \beta_{\text{survivor}} < \bar{\beta}_p.$$

Since the model implies $\bar{\beta}_a \in [0, \beta_{\text{survivor}}]$ and $\bar{\beta}_p \in [\beta_{\text{survivor}}, 1]$, an application of Proposition 5 yields

$$\beta_{\text{cohort}} \in \left[(1 - \theta^a) \beta_{\text{survivor}}, (1 - \theta^p) \beta_{\text{survivor}} + \theta^p \right].$$

This is tighter than worst-case bounds but does not sign $\beta_{\text{survivor}} - \beta_{\text{cohort}}$; the direction depends on whether the upward contribution of avoidance exceeds the downward contribution of participation exit. If

$$\theta^a (\beta_{\text{survivor}} - \bar{\beta}_a) > \theta^p (\bar{\beta}_p - \beta_{\text{survivor}}), \quad (19)$$

then measured passthrough is overstated; $\beta_{\text{cohort}} < \beta_{\text{survivor}}$. Without knowing average passthrough of exiters, we can still construct a dominance condition using the model. Letting $\bar{\beta}_p = 1$ take its worst-case value, transshipment dominates and $\beta_{\text{cohort}} < \beta_{\text{survivor}}$ whenever

$$\bar{\beta}_a < \beta_{\text{survivor}} - \frac{\theta^p}{\theta^a} (1 - \beta_{\text{survivor}}).$$

Given measurements of θ^a and θ^p and measured passthrough β_{survivor} (typically estimated near one), we need only be confident that $\bar{\beta}_a$ is less than the combination of them from the right-hand side of the inequality above to ensure that $\beta_{\text{cohort}} < \beta_{\text{survivor}}$. The following section develops a methodology to measure each margin.

4 Constructing Exit Measures in Customs Data

The model shows that the 2018 tariff war gave Chinese exporters an incentive to either cease trade with the U.S. or to transship goods through a third country. The resulting measured passthrough of tariffs onto buyer prices β_{survivor} may be biased relative to β_{cohort} since it is typically measured with bilateral trade data, which conditions on the survivor set. This section discusses how to measure both exit margins in customs data in three steps. I first describe the customs data and tariff wedge, then map the firm-level objects from the model to product-level measures, and finally construct the two exit margins.

4.1 Data

The ideal dataset would contain firm-level exporter data from China, tracing their trade across countries together with the associated tariff wedges. Instead, it is more useful to illustrate the framework using the datasets typically available to researchers investigating the passthrough of tariffs onto buyer prices. In particular, I use annual data on bilateral trade flows from CEPII's BACI database (HS2012 classification, V202501 release) from 2012–2023 at the 6-digit product (HS6)

level.¹² Amiti, Redding, and Weinstein (2019), Amiti, Redding, and Weinstein (2020), Fajgelbaum et al. (2020), and parts of Cavallo et al. (2021) all use bilateral product-level trade data to estimate a passthrough around one.

For each year t , country pair (i, j) , and HS6 code k , I observe the customs value of shipments $v_{i \rightarrow j, k, t}$. I use these flows to measure both forms of exit from direct China–U.S. trade. I discuss each measure in more detail later in this section. I identify patterns consistent with transshipment by looking for corresponding changes in China-to-hub and hub-to-U.S. trade flows for the same product. I detect participation by looking for persistent declines in direct trade that are not accounted for by detected transshipment. Both measures noisily capture true participation and transshipment. I restrict potential hubs to those identified by the White House Council of Economic Advisers (CEA 2026) as transshipment hubs, as well as additional countries selected for geographic proximity to China, port infrastructure, and pre-tariff trade relationships with both the U.S. and China like Hong Kong and Macau. This results in a sample of 73 countries. I deflate all values to 2017 dollars using the BLS end-use import price index.

Since the later regressions are at the product-year level, I construct the tariff wedge at that level. Following the model, let $\tau_{k,t}^d$ denote the direct tariff on product k at time t for a good imported from China to the United States. Moreover, let $\tau_{k,t}^h$ denote the average U.S. tariff on product k imported from candidate hub countries in year t , weighted by their 2017 shares of U.S. imports of that product. This is analogous to τ^a in the model, but τ^h incorporates only statutory tariffs, not the additional costs of transshipment. I summarize the tariff incentive to transship by defining the wedge as

$$\Delta_{k,t} \equiv \log \frac{1 + \tau_{k,t}^d}{1 + \tau_{k,t}^h}.$$

If this expression is equal to zero, then there is no tariff advantage relative to the weighted-average hub, although individual hubs may still offer lower tariffs. If it is greater than zero, then the U.S. tariff on Chinese goods exceeds the weighted-average hub tariff, creating a tariff incentive to transship before accounting for other costs. Unlike in the model, the wedge can vary across products; I extend the model to incorporate that in Supplemental Appendix S.1.4. For the direct tariff, I use Amiti, Redding, and Weinstein (2020) for monthly HS10 China-specific rates in 2016–2017, aggregated to annual HS6 using 2017 import shares as weights. I hold the 2016 rates fixed for 2012–2015. From 2018–2023, I use the Global Tariff Database from Rodríguez-Clare, Ulate, and Vasquez (2025), which provides bilateral HS6 tariffs based on the methodology in Teti (2025). For the hub tariff $\tau_{k,t}^h$, I use the U.S. tariff faced by the candidate hub countries on each HS6 product,

12. BACI provides consistent bilateral flows at annual frequency with comprehensive country coverage. The HS6 level is the finest internationally harmonized classification in bilateral trade data; finer classifications are country-specific.

drawn from the same Global Tariff Database and value-weighted across hubs by their pre-tariff shares of U.S. imports.

The average increase in the wedge following the trade war is roughly 0.100. It is useful to break this down by end-use category. I classify each HS6 product as consumption, intermediate, or capital goods using the UN’s Broad Economic Categories (BEC). Table C.1 reports the resulting wedge by end-use category, assigning each product to its dominant 2017 end use. In these value-weighted summary statistics, the post-minus-pre increase in $\Delta_{k,t}$ is largest for intermediate goods (0.167), followed by capital goods (0.140) and consumption goods (0.054). The differences across end-use categories reflect the underlying direct and hub tariffs, with intermediates facing the highest direct rates.

4.2 From Firms to Products

It is important to map the data into the model to interpret what any resulting estimates may mean economically. This is not straightforward because, whereas the data observe products as HS6×origin cells, the model describes choices by individual firms which produce a particular variety. One plausible interpretation is of the model’s firm as a variety within product k , which may contain many such varieties. This interpretation of the data–model mapping connects the model to what passthrough regressions typically estimate. In such regressions, researchers estimate the average passthrough of tariffs to buyer prices using tariff changes within product–origin cells. If we interpret varieties as lying within these cells, then we can use the framework to correct within-product passthrough for selection into exit at the same level of aggregation.

Implementing the correction requires measuring the share of the original trade base associated with each exit margin. While I turn to exactly how I do that in the next subsection, it is useful to first define it conceptually. Toward that end, let θ_k^a denote the share of product k ’s baseline direct-trade value associated with varieties that subsequently transship, and let θ_k^p denote the corresponding share associated with varieties that cease serving the U.S. market. Aggregating across products with 2017 China–U.S. import shares as weights ω_k gives

$$\theta^r = \sum_k \omega_k \theta_k^r, \quad r \in \{a, p\}.$$

The shares θ^a and θ^p measure how much of the original China–U.S. trade base is associated with each exit margin, accounting for differences in both product size and product-level exit. Combining these shares with survivor passthrough and restrictions on passthrough among exiting activity allows the framework to correct the pooled passthrough coefficient. To translate this correction into buyer-side incidence, the aggregation must also account for differences in tariff

exposure. For that calculation, both exit shares and passthrough are aggregated using weights proportional to each product’s 2017 China–U.S. import value multiplied by its direct tariff-rate increase, as described in Section 2.3. Denoting the exposure-adjusted shares as $\tilde{\theta}^r$, the resulting correction is

$$\beta_{\text{cohort}} = (1 - \tilde{\theta}^a - \tilde{\theta}^p)\beta_{\text{survivor}} + \tilde{\theta}^a \bar{\beta}_a + \tilde{\theta}^p \bar{\beta}_p,$$

where $\bar{\beta}_r$ is the average exiter passthrough within each channel.

Measuring these exit shares is complicated by the aggregation of varieties within products. A product may continue to trade after some varieties exit, while a decline in its trade may also reflect lower sales by varieties that remain. The following subsections use patterns in direct and third-country trade to construct product-level proxies for the transshipment and participation shares entering this correction.

4.3 Constructing the Exit Share

I measure the components of the exit share—transshipment θ^a and participation θ^p —separately and with the intention of only capturing tariff-induced exit. Neither component is directly observed in these data, so I use an approach common in the trade literature of proxying for these margins using circumstantial evidence (Fisman and Wei 2004; Fisman, Moustakerski, and Wei 2008; Freund 2024). Since these methods are plausibly sensitive to supply chain reallocation that happened as a result of the Covid-19 pandemic, I restrict the initial selection of activity to 2018–19, which is the period most passthrough studies analyze, and follow that activity in subsequent years. I discuss each measure in turn.

4.3.1 The Transshipped Share

Detecting transshipment is inherently difficult because bilateral customs data cannot directly reveal the origins of any particular shipments. If they could, then it would not be a struggle for the United States government to punish transshipment, and researchers could simply adjust for the affected shipments when running passthrough regressions. Given that difficulty, I develop low and high measures of the transshipment share θ^a by screening customs data for patterns consistent with Chinese goods flowing through third countries en route to the United States. My approach extends Freund (2024), who develops conditions to identify transshipment in bilateral trade data. Using bilateral data allows me to examine these patterns across a broad set of countries, complementing firm-level evidence from Vietnamese customs records (Iyoha et al. 2025) and bill-of-lading data (Do et al. 2025). The exact details of the implementation are in Appendix B; I

discuss them broadly below.¹³

My approach to identifying transshipment requires that trade follow a particular pattern. It must be that, for some product, China loses U.S. import share, a hub gains share in the same product, and China's shipments to that hub rise above their pre-trend. I evaluate all trade shares and trade flows against pre-trends estimated from 2012–2017.

1. The product must have faced U.S. tariff increases on China, which restricts attention to goods with a policy-induced incentive to transship.
2. The hub must face a lower U.S. tariff than China on that specific product, ensuring an incentive to transship through that hub. Tariff exposure is evaluated in the year being measured. For the direct-versus-hub tariff comparison, I use the corresponding year's rates in 2018–2019 and hold them at their 2019 values thereafter, which is generally innocuous since there were few changes after 2019.
3. China's share of U.S. imports must have fallen below its pre-trend while the hub's share rose above its own, a reallocation pattern consistent with transshipment.
4. The hub's deviation from its pre-trend market share in non-U.S. markets must be no greater than China's corresponding deviation. Comparing China and the hub outside the United States helps distinguish U.S.-specific reallocation from a broader shift in competitiveness.
5. Both China-to-hub and hub-to-U.S. trade must exceed their pre-trends, with the smaller observed flow at least \$200,000 in current dollars.

I implement these screens together with an additional persistence requirement. For example, to flag a route in 2019, conditions 3–5 must hold in 2019 and in either 2018 or 2020; otherwise, the route is not flagged. That is to account for the fact that these routes are costly to set up and it is unlikely that a firm would want to transship in one year but not another. For 2020–2023, I follow only products detected in 2018 or 2019, while reassessing the qualifying routes and their trade flows each year. This focuses the later estimates on products with evidence of transshipment during the initial tariff increases.

Once a route satisfies these conditions, I determine how much of its trade to attribute to transshipment. I use two different methods to do this. The first, which I call the *low screen*, assigns the smaller above-trend increase across the China-to-hub and hub-to-U.S. flows, while the *high screen* assigns the smaller observed trade value. The two measures use the same qualifying routes but differ in the amount attributed to transshipment. For example, suppose Chinese trade to a

13. The measures focus on transshipment and omit other avoidance margins, such as value misreporting and product misclassification. They therefore do not measure total avoidance and they are not intended to.

third country is \$10 million for a particular product, and the corresponding trade for that product to the United States is \$8 million. Further, suppose that this trade is associated with above-trend increases of \$3 million and \$2 million, respectively. Under the low screen, \$2 million would be associated with transshipment, while the high screen would attribute \$8 million to transshipment. I sum these amounts across qualifying hubs within each product. I apply product-level caps and express the resulting amounts in 2017 dollars, so that measured transshipment cannot exceed a product’s baseline direct-trade value. I then sum the resulting amounts across products and divide by the total 2017 direct China–U.S. trade value of the tariff-exposed sample to obtain θ^a .

To assess potential false positives, I apply the trade-pattern and persistence conditions to products that did not face the 2018–19 tariff increases. Over 2018–2023, even the high measure attributes no more than 1.36 percent of this group’s 2017 direct China–U.S. trade value to transshipment in any year. These relatively small placebo shares support the view that the main estimates are not simply capturing ordinary changes in bilateral trade.

External Validation. My procedure for constructing transshipment estimates is largely validated as a conservative approach by other sources. For example, Iyoha et al. (2025) use transactions data to show that transshipment accounts for 8.8 percent of Vietnam’s growth in U.S.-destined exports between 2018 and 2021. Over the same window, my low and high screens attribute 4.9 and 7.9 percent of that export growth to transshipment, respectively.¹⁴ For 2023, Goldman Sachs estimated approximately \$40 billion in transshipment, which is slightly higher than my high screen, while other estimates from the White House Council of Economic Advisers, the Department of Commerce, and other sources are far more aggressive (CEA 2026). Altogether, the transshipment estimates tend to be more conservative than those from other sources.

4.3.2 The Participation Share

The model’s version of participation exit consists of exporters which cannot profitably serve the U.S. market, so they cease to trade at all with the U.S. Clearly, this is unobservable in product-level data, and so, as with transshipment, I construct a proxy for it. In addition to the HS6 data, I use monthly HTS10 data from the Census Bureau, linking codes across classification changes over time. These more granular data are a useful supplement because they identify which detailed categories account for contractions within an HS6 product. They cannot, however, be used for the same cross-country matching of transshipment routes because the more granular codes are not harmonized across countries. The remainder of this section provides an overview of how I combine those two datasets to screen for participation exit, while details are in Appendix B.8.

14. This comparison comes from allowing my screens to include products first detected beyond 2019. When restricted to products that show evidence of transshipment in 2018–19, the corresponding range is 3.4%–5.2%.

I proxy for participation exit by looking for disappearance or persistent contractions in direct China–U.S. trade which do not feature evidence of substantial transshipment. For each HTS10 category ℓ , let $r_{\ell,t}$ be actual real China–U.S. trade divided by the smaller of its 2012–2017 log-linear pre-trend forecast and its actual real trade in 2017. As a first step, a category gets flagged for participation exit if it records no direct trade in December 2019, or if $r_{\ell,t} < \epsilon_p^*$ in both 2018 and 2019. I choose the contraction threshold ϵ_p^* to limit false positives using the pre-tariff period. I calibrate the cutoff using the comparison with predicted trade, then apply the same cutoff to the smaller of predicted trade and actual 2017 trade. For the calibration, the forecast comparison must flag categories representing no more than 1 percent of the sample’s 2017 trade between the U.S. and China. This limits how much trade a particular proposed cutoff would mistakenly classify as tariff-induced participation exit. For example, I predict each HTS10 category’s trade in 2016 and 2017 using its 2012–2015 trend. I then apply the proposed cutoff to the ratio of actual to predicted trade in each year. Categories falling below the cutoff in both years, but still trading in December 2017, count as placebo false positives, whose combined 2017 baseline trade value I express as a share of the cohort’s total 2017 direct-trade value. I repeat this exercise for every pair of consecutive years within 2012–2017, estimating each trend using the remaining years, and select the largest threshold that keeps this share at or below 1 percent in every comparison. This yields $\epsilon_p^* = 0.273$. I require direct trade in both 2018 and 2019 to remain below 27.3 percent of the smaller of its predicted level and its actual 2017 level. The comparison with actual 2017 trade prevents slower-than-predicted growth alone from qualifying as participation exit.

As a second step, I take all HTS10 categories flagged under the first screen and exclude those whose HS6 product has substantial measured transshipment in both 2018 and 2019. This helps distinguish participation exit from measured transshipment. I calibrate how much measured transshipment is sufficient to exclude a product from the participation screen using the same pre-tariff two-year windows and identifying products whose measured transshipment shares reach a candidate threshold in both years. I calibrate the threshold so that these products account for no more than 1 percent of the cohort’s 2017 direct-trade value in every window. Applying this procedure separately to the low and high transshipment screens gives exclusion thresholds of 11.24 and 26.78 percent, respectively. For the years after 2019, I follow only HTS10 categories flagged in 2018–19 and reassess participation each year. For years prior to 2018, I report the corresponding placebo measures.

If HTS10 categories pass both stages, they qualify as participation exit. In that case, I quantify this margin using their combined share of each HS6 product’s 2017 Census trade, multiplied by that product’s 2017 trade value in BACI. For example, if an HS6 product accounted for 0.5 percent of the sample’s direct China–U.S. trade in 2017, and HTS10 categories representing 20 percent of that product’s baseline trade passed both screens, it would contribute 0.001 to the aggregate

participation share θ^p .

5 Measured Exit in Customs Data

With the two exit margins constructed, I now estimate the magnitude of each exit margin and provide evidence on the direction of selection.

5.1 The Magnitude of Exit

I start by showing the magnitude of exit in levels. The top left panel of Figure 2 shows transshipment and participation from 2012 through 2023, each measured relative to the original 2017 trade base. Before the trade war, my constructed measures pick up very little transshipment or participation. After tariffs picked up in 2018, transshipment rises to between 2.6 and 7.3 percent of baseline trade in 2019, depending on the screen. It continues to rise through 2021–2022 before declining to between 4.8 and 8.4 percent in 2023. Participation follows a different pattern. It reaches approximately 1 percent in 2019, then falls to about 0.45 percent by 2023. Thus, most measured exit from direct trade takes the form of transshipment rather than firms ceasing trade with the United States altogether.

The remaining panels break down each exit margin by end-use. The differences between each are substantial. Whereas just 2.6 percent of consumption goods trade were transshipped in 2023, 13.2 percent of intermediates and 10 percent of capital goods were. The same order applies under the low screen. Similarly, participation exit is largest among intermediates, accounting for roughly one percent of baseline trade in 2023, compared with roughly 0.1 percent for capital and consumption goods. These differences indicate that exit was largest among business inputs, which is sensible given that these inputs also had the largest tariff increases.

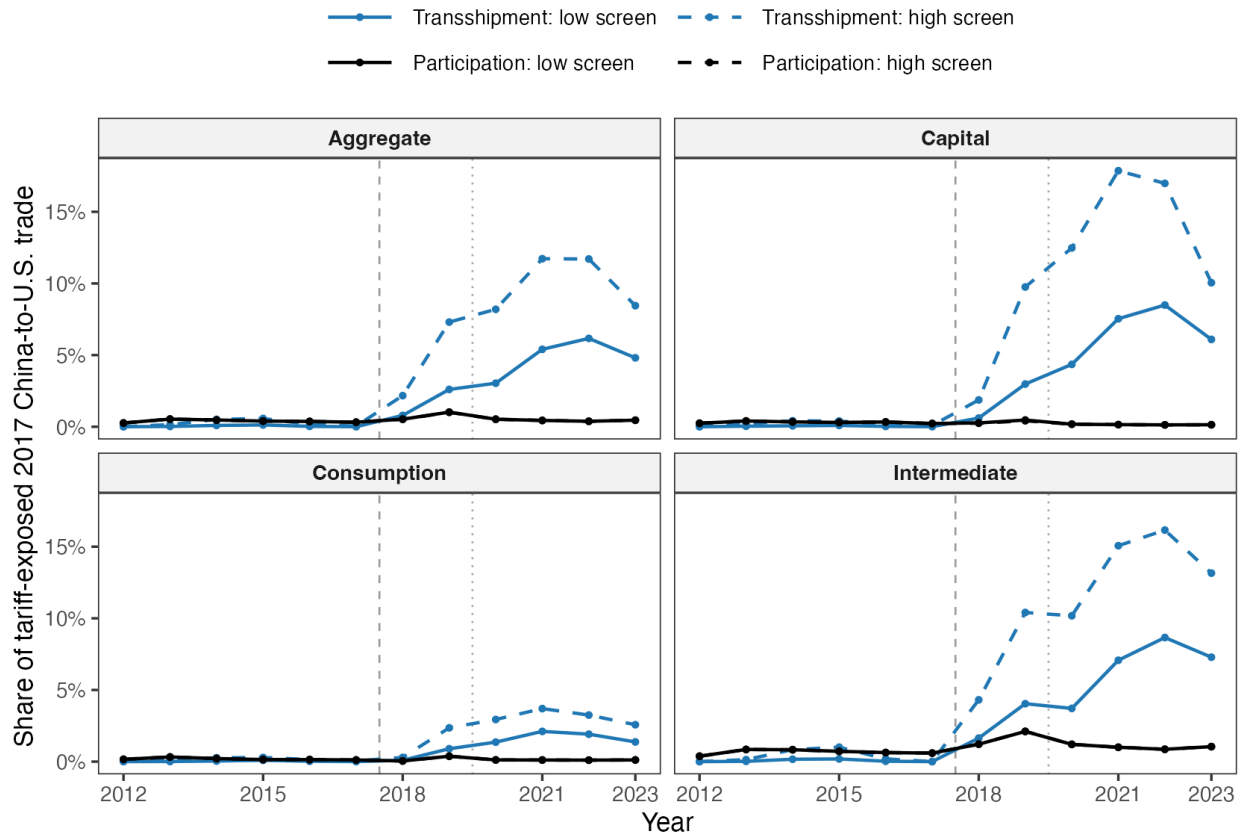
I now turn to asking whether regression evidence validates what we observe in levels. In the model, a larger gap between the direct and hub tariffs raises the gain from transshipment, allowing more exporters to justify its fixed cost. Consequently, I ask whether products facing larger increases in the tariff gap are associated with larger increases in transshipment. To test that, I estimate

$$\theta_{k,t}^a = \alpha_k + \delta_{t \times e} + \gamma \Delta_{k,t} + \varepsilon_{k,e,t}, \quad (20)$$

where e indexes end use, α_k are HS6 product fixed effects, and $\delta_{t \times e}$ are year-by-end-use fixed effects. The tariff wedge $\Delta_{k,t}$ measures the difference between the log gross direct and hub tariffs. I use annual observations from 2012–2023, weight by 2017 China–U.S. import values, and cluster standard errors by product. Product fixed effects account for any permanent differences between products, while year-by-end-use fixed effects account for shocks common to end-use categories

within a given year. To interpret γ as a tariff-induced response, we must assume that products facing different wedge increases would not otherwise have experienced systematically different changes in transshipment after accounting for the fixed effects. I also estimate a corresponding regression with participation exit on the left-hand side, and the direct tariff on the right-hand side.

Figure 2: Transshipment and Participation Exit Shares



Notes: Transshipment (θ^a) and participation (θ^p) as shares of 2017 direct China–U.S. trade among tariff-exposed products. Panels report the aggregate and each end-use category. Solid lines denote the low screen; dashed lines denote the high screen. Pre-2018 observations are placebo measures.

Table 1 reports the results for several specifications. Column 1 reports the coefficients from the low screen estimated separately for aggregate transshipment and each end use category; Column 2 does the same for the high screen. In both specifications, θ is a continuous share, and the regressions include observations with zero measured exit. By contrast, Column 3 uses an indicator for any detected transshipment, capturing the extensive margin. I find that the aggregate semi-elasticity is 0.236 under the low screen and 0.474 under the high screen, and both are statistically significant at the 1 percent level. To interpret the magnitudes, suppose that the direct tariff on a product rises from zero to ten percent. This would correspond to an increase

in transshipment of roughly 2.25 percentage points under the low screen and 4.52 percentage points under the high screen. Mimicking the results from the figure, transshipment responds positively and significantly for every end-use category, but business inputs are most responsive, while consumption goods are considerably less so. On the extensive margin, only intermediates show a response significant at the 1 percent level, while the aggregate coefficient and the other end uses do not. On the participation side, Columns 4 and 5 show no statistically significant response of participation exit to tariffs, so these regressions do not provide evidence of the model's predicted positive response.

Next, I examine whether the measured response of transshipment is consistent with the model's mechanism. In principle, exporters benefit from rerouting only when tariff savings cover the additional costs, implying more transshipment for products facing lower U.S. tariffs when imported through hub countries, holding direct tariffs and other route costs fixed. I examine this prediction in Appendix Table C.2, comparing post-2017 changes in transshipment across products facing different hub tariffs. The relationship is negative under both screens and statistically significant under the high screen, which provides moderate evidence in favor of the prediction.¹⁵ Beyond hub tariffs, exporters must also cover the costs of transporting goods through an additional country. For example, additional shipping costs should absorb more of the tariff savings for bulky goods than for high-value-to-weight goods such as electronics. Consequently, the transshipment response should be weaker for bulkier products and stronger for electronics. In Supplemental Table S13, I interact the tariff wedge with pre-tariff product characteristics and find that the transshipment-share response is weaker for bulkier products and stronger for electronics. These interactions are statistically significant at the 10 and 5 percent levels, respectively, under both screens.¹⁶ These patterns support the interpretation that measured rerouting responds to the net savings from avoiding direct tariffs.

Several robustness and falsification tests assess alternative explanations for the estimated response of transshipment. I first investigate whether products may have differential pre-trends, which is the main threat to interpreting the baseline relationship as a response to tariffs. It could be that products which later received larger wedge increases may already have been moving toward more transshipment. However, the event study in Supplemental Figure S2 shows that the large transshipment-share responses emerge after the tariffs. Next, I check whether products that were later hit with higher tariffs already had different measured transshipment prior to the

15. The selection mechanism predicts lower survivor passthrough when higher hub tariffs retain more low-passthrough exporters in direct trade. The annual BACI results run against this prediction: higher hub tariffs are associated with a significantly larger price response to increases in direct tariffs. The corresponding interactions are statistically insignificant in monthly and fixed-basket Census data (Supplemental Table S15).

16. The interaction with pre-tariff hub intensity is positive but not statistically significant. Supplemental Table S14 reports the corresponding regressions for any detected transshipment; none of these interactions is statistically significant.

trade war. I compare their average pre-tariff transshipment with that of products that did not receive these increases. I then randomly reassign which products belong to each group within HS4 categories 5,000 times, preserving each category's number of treated products, and repeat the comparison to assess how unusual the actual difference is in Supplemental Table S2. Products that subsequently received tariffs had slightly lower, rather than higher, measured transshipment before 2018. The two-sided permutation p -values are 0.563 under the low screen and 0.446 under the high screen. Relatedly, I examine whether the positive transshipment response depends on pre-tariff imbalance or product-specific trends. Using entropy balancing, HS4 trends, residualizing HS6 pre-trends, long differences, and a 2016–2019 window all preserve positive and statistically significant transshipment-share coefficients in Supplemental Tables S12, S11, S9, S10, and S8, respectively.

The preceding permutation test compares pre-tariff transshipment between products that later received tariff increases and products that did not. I use a second permutation test to ask whether the estimated relationship between tariff changes and transshipment over the full sample could arise by chance. In Supplemental Figure S3, I randomly reassign full tariff-wedge trajectories across products 5,000 times and re-estimate the main specification, keeping measured transshipment and regression weights unchanged. This asks whether similarly strong relationships appear when products are assigned tariff histories they did not actually face. Almost none of these random assignments produces a relationship as strong as the actual one. I also test whether the screens capture a global reorientation of Chinese exports after 2018 and are not necessarily the result of U.S. tariffs. To do that, I apply the same screening criteria to two placebo destinations which did not impose the same tariff increases on Chinese goods: the EU and UK on the one hand, and Canada on the other. In Supplemental Table S3, I find that, after 2018, measured transshipment responses are small and largely insignificant for both economic areas.

The remaining tests relate to specification choices about the baseline regression. I find that clustering by HS4 and year (Supplemental Table S4), different fixed effects (Supplemental Table S5), using an unweighted regression (Supplemental Table S6), and using the direct tariff alone (Supplemental Table S7) all preserve positive and statistically significant transshipment-share responses by end use and in the aggregate. Finally, I examine whether the result is carried by a single hub. I reconstruct the measures and re-estimate the regressions after excluding each of the 73 hub economies in turn. The transshipment-share response remains positive and statistically significant under every exclusion (Supplemental Figure S4).

Table 1: Tariff Responses of Transshipment and Participation

	Transshipment			Participation	
	Low	High	Any	Low	High
	screen	screen		screen	screen
	(1)	(2)	(3)	(4)	(5)
Aggregate	0.236*** (0.054)	0.474*** (0.112)	1.372 (0.961)	0.000 (0.019)	0.000 (0.019)
Capital	0.320*** (0.103)	0.686*** (0.203)	1.533 (1.435)	−0.001 (0.006)	−0.002 (0.006)
Consumption	0.144*** (0.034)	0.284*** (0.067)	1.143 (0.980)	0.001 (0.004)	0.001 (0.004)
Intermediate	0.249*** (0.066)	0.429*** (0.137)	1.492*** (0.311)	0.002 (0.083)	0.002 (0.083)
Observations	52,872	52,872	52,872	52,872	52,872
HS6 products	4,081	4,081	4,081	4,081	4,081
HS6 fixed effects	✓	✓	✓	✓	✓
Year × end-use FE	✓	✓	✓	✓	✓
2017 import-value weights	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Annual observations, 2012–2023. Columns (1)–(2) use transshipment shares; column (3) uses an indicator for any transshipment. Their regressor is the change in the log gross direct-to-hub tariff wedge. Columns (4)–(5) use participation shares and the change in the log gross direct tariff. Shares and indicators are on a 0–1 scale. Aggregate and end-use coefficients come from separate specifications. Standard errors clustered by HS6 are in parentheses.

5.2 Selection Across Exit Margins

The model makes distinct predictions about the characteristics of exiters along each margin. On the one hand, low-cost exporters have greater sales and lower passthrough, and gain more from paying the fixed cost to transship goods. On the other, high-cost exporters have smaller sales and

higher passthrough, and are more likely to cease trading with the U.S. market. Table 2 studies these patterns across products by separating them into four groups: products with evidence of transshipment alone, products with evidence of participation exit alone, products with evidence of both margins, or neither during 2018–19.

I compare these groups' sales and China's share of U.S. imports before the tariffs, using observations from 2012–2016. I estimate a PPML regression of average sales and an OLS regression of China's standardized market share on indicators for belonging to each exit group, controlling for tariff exposure and other pre-tariff characteristics along with broader HS4 product fixed effects. Products with neither exit margin within an HS4 group are the comparison group. The first two columns compare the sales of each group. Relative to products with neither exit margin, transshipment-only products had pre-tariff sales approximately 22 percent higher, while participation-only products had sales approximately 12 percent lower. The transshipment-only difference is statistically significant, whereas the participation-only difference is not. Transshipment-only products had sales approximately 39 percent higher than participation-only products, a statistically significant difference under both screens. The model's sales ordering reflects differences in exporters' costs within a common demand environment. Across products, however, larger Chinese sales may simply reflect a larger U.S. import market. Consequently, I also examine import shares to assess whether the sales differences are accompanied by differences in China's position relative to competing suppliers. China's pre-tariff import share was approximately 0.22 standard deviations higher for transshipment-only products than for participation-only products, a statistically significant difference under both screens. Thus, compared with transshipment-only products, participation-only products began with both smaller Chinese sales and a weaker Chinese position relative to other supplying countries.¹⁷

The model makes a prediction about prices, namely that the higher costs of participation exiters imply smaller sales and higher prices. I use the more granular HTS10 data to study this. Appendix Table C.4 examines this prediction by comparing the 2017 characteristics of HTS10 categories subsequently flagged for participation exit with those of unflagged categories within the same HS6 product and physical unit of measurement. Under equal-category weighting, participation exiters had unit values approximately 0.27 log points higher, together with substantially lower quantities and sales; all three differences are statistically significant and accord with the model's predictions. While the quantity and sales differences remain statistically significant under trade-value weighting, the unit-value difference remains positive but is imprecisely estimated. This high-price, low-sales profile is consistent with participation exit coming from higher-cost va-

17. Appendix Table C.3 examines the cross-product comparisons after controlling for pre-tariff trade connections and hub identities, and under 2017 trade-value weighting. The sales difference remains positive and statistically significant at the 5 percent level with the additional routing controls and at the 10 percent level under trade-value weighting.

rieties within a product. Since transshipment is identified at the HS6 level rather than HTS10, the same test cannot be run for transshipment, so I run an analogous cross-sectional test across products in Appendix Table C.5. I do not find a significant relationship between pre-tariff Chinese unit values and transshipment across HS6 products.¹⁸

Table 2: Pre-tariff Characteristics by Subsequent Exit Margin

Product group	Pre-tariff sales (PPML)		China's pre-tariff U.S. import share (standard deviations)	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
Transshipment only	0.201*** (0.076)	0.200*** (0.076)	0.069 (0.044)	0.064 (0.043)
Participation only	-0.127 (0.101)	-0.126 (0.101)	-0.155*** (0.038)	-0.154*** (0.038)
Transshipment and participation	0.381*** (0.145)	0.386*** (0.148)	0.070 (0.056)	0.084 (0.058)
Transshipment minus participation	0.327*** (0.101)	0.326*** (0.101)	0.223*** (0.052)	0.218*** (0.051)
Products	3,934	3,934	3,934	3,934

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Outcomes are 2012–2016 averages. Sales columns report PPML coefficients; import-share columns report OLS coefficients. The omitted group has neither exit margin. All specifications use equal product weights, HS4 fixed effects, and controls for tariff exposure, end use, routing, bulkiness, concentration, and route capacity. HS4-clustered standard errors are in parentheses.

Taken together, the pre-tariff sales comparisons and the within-product participation evidence support the model's prediction of opposing selection across exit margins. Which margin dominates depends on both its size and its passthrough difference from survivors. The next section combines the measured exit shares with restrictions on passthrough among exiting activity to bound the resulting correction.

18. The separate measure of China's price premium relative to world unit values is positively associated with transshipment among non-differentiated products.

6 Corrected Tariff Passthrough and Incidence

Having established the magnitude of exit and its likely agreement with the model, I use the results of Section 2 to bound cohort passthrough and the incidence attributed to the original import base. As an illustrative exercise, I use the exit shares measured in 2023 and set $\beta_{\text{survivor}} = 1$, consistent with the near-complete passthrough estimated in studies using bilateral trade data (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020; Cavallo et al. 2021). Moreover, I reweight products by their 2017 direct China–U.S. trade value multiplied by their completed 2018–19 direct tariff-rate increase, as described in Section 4.2. Under these exposure-adjusted weights, the avoidance shares are 5.67 and 9.83 percent under the low and high screens, respectively, while participation accounts for approximately 0.50 percent under both. With that information, Table 3 reports bounds and calibrated corrections to survivor passthrough, which I walk through subsequently.

The first numeric column reports worst-case bounds for passthrough. Given the information in the previous paragraph and the model-supplied bounds on passthrough of $[0, 1]$, that column reports aggregate worst-case bounds of $[0.897, 1]$ spanning the low and high screens. The worst-case bounds to consumption are narrow, while they are wider than the aggregate for business inputs. The second numeric column then tightens the bounds by imposing $\bar{\beta}_a \leq \beta_{\text{survivor}} \leq \bar{\beta}_p$ on the pooled exiter averages, applying the selection ordering developed in the model. Since participation exiters have passthrough at least as high as survivors, their passthrough must equal one at the unit survivor anchor. However, these corrections are minor.

While the first two columns made the worst-case assumption on exiter passthrough using the model-supplied bounds, the third and fourth columns use an external calibration of exiter passthrough to calculate calibrated passthrough. Using evidence on the relationship between firm size and passthrough from Amiti, Itskhoki, and Konings (2019), I calibrate the passthrough of transshippers to $\bar{\beta}_a = 0.525$, while setting $\bar{\beta}_p = 1$, the highest participation-exiter passthrough permitted by the model and the value least favorable to avoidance dominance. Given those assumptions and the exit shares, the third column reports how large survivor passthrough would have to be for transshipment exit, which is on the low-passthrough side of the distribution, to dominate participation exit, which pulls from the high-passthrough side, under both screens. I find that the passthrough of survivors must exceed a threshold of around 0.55 across end uses, well below typical estimates for the trade war. With that, the fourth column supplies a range of calibrated passthrough across screens for each end use. The calibrated aggregate cohort passthrough is 0.953–0.973, a reduction of approximately 2.7–4.7 percentage points from the unit survivor anchor. The correction is smallest for consumption goods, whose calibrated passthrough is 0.986–0.993, compared with 0.948–0.968 for capital goods and 0.934–0.963 for intermediates. Follow-

ing Section 2.3, under competitive domestic production these ranges also describe the combined first-order incidence attributed to domestic firms and consumers in the original import base. The burden is expressed per dollar of additional tariff payments calculated holding 2017 trade values fixed.

Table 3: Bounds, Dominance, and Cohort Passthrough (2023)

End use	Worst-case bound	Direction-only bound	Dominance threshold	Calibrated passthrough
Aggregate	[0.897, 1]	[0.902, 1]	0.564	0.953–0.973
Capital	[0.889, 1]	[0.890, 1]	0.536	0.948–0.968
Consumption	[0.970, 1]	[0.971, 1]	0.568	0.986–0.993
Intermediate	[0.851, 1]	[0.861, 1]	0.578	0.934–0.963

Notes: Cohort passthrough is evaluated at $\beta_{\text{survivor}} = 1$, with support $[0, 1]$. The worst-case bound uses the total exit share; the direction-only bound additionally imposes $\bar{\beta}_a \leq \beta_{\text{survivor}} \leq \bar{\beta}_p$. Bounds span both screens. Dominance thresholds are the maximum across screens; calibrated passthrough reports the range. Both use $\bar{\beta}_a = 0.525$ and $\bar{\beta}_p = 1$. Weights are proportional to 2017 direct China–U.S. trade value multiplied by the completed 2018–19 Section 301 tariff-rate increase.

The magnitude of the correction is robustly modest when accounting for different concerns. For example, one may worry that the assumption on the passthrough of transshippers is too small. The correction is linear in that, and Appendix Table D.2 varies it for different estimates from Amiti, Itskhoki, and Konings (2019). At a very high passthrough of $\bar{\beta}_a = 0.9$, the correction falls to 0.57–0.98 percentage points. Next, one may also worry, based on recent work from Alvarez et al. (2023), that β_{survivor} ought to be smaller than one. Using continuing-relationship trade, one can infer from their estimates that passthrough is roughly 0.8. In that case, retaining $\bar{\beta}_a = 0.525$ and $\bar{\beta}_p = 1$ implies a correction of between roughly 1.5 and 3.5 percentage points. Third, there may be a countervailing compositional effect: tariffs could induce importers to shift toward cheaper goods, which could offset the overstatement caused by selection. Appendix Table D.1 separates changes in unit values from changes in the mix of goods traded. I find that, in the matched 2017–2019 sample and under the central calibration, compositional effects offset selection when the fraction of the measured composition effect that remains in the passthrough coefficient after controlling for other factors reaches 10.8 percent under the low screen or 25.8 percent under the high screen. In other words, even if the regression accounts for most of the composition effect, the remainder can still be large enough to offset the overstatement of cohort passthrough caused by selection. Nevertheless, that is not the focus of this paper.

In their study of the trade war, Alvarez et al. (2023) find lower passthrough among continuing buyer–supplier relationships than in broader observed samples, in which estimates approach one. Their finding is relevant to the survivor–cohort comparison because relationships can disappear when exporters leave direct trade. Restricting attention to continuing relationships can therefore capture some of the selection studied here, and their result raises the possibility that selection lowers measured passthrough. However, two ambiguities complicate that interpretation. First, a relationship can end while the exporter continues selling directly to other U.S. buyers. Their restriction therefore excludes both activity that leaves direct trade and transactions that remain in the tax base but do not belong to continuing relationships. Second, their sample may retain transshippers while excluding exporters who entirely cease trading with the United States. Under the model’s selection ordering, this pattern would retain low-passthrough activity while removing high-passthrough activity, selecting the observed sample toward lower passthrough. Thus, their findings provide relevant evidence about selection, but the implication for the sign of the survivor–cohort gap remains ambiguous.

7 Conclusion

This paper generalizes the canonical textbook approach to tax incidence to environments in which passthrough is heterogeneous and agents have the option to exit the observed tax base. In such cases, measured passthrough can be biased, and the direction of the bias depends on the sign of the covariance between passthrough and survival. This insight applies across many settings. For example, consider top personal income tax rates. When they rise, top earners can respond to high marginal rates by moving to lower-tax jurisdictions, which removes them from the observed tax base and can bias incidence inferred from those who remain if movers differ systematically in passthrough (Kleven et al. 2020; Rauh and Shyu 2024). The same logic applies in lower-income settings, where small firms and low-skill workers may move into the informal sector to escape the tax (Ulyssea 2020). Corporate income taxes create another version of the same problem: firms can shift paper profits abroad (Tørsløv, Wier, and Zucman 2023) or relocate real activity (Giroud and Rauh 2019). The correction is general, while its incidence interpretation depends on the tax instrument and market structure.

I apply the insight to the 2018 trade war between China and the United States. Revisiting literature estimates that passthrough of tariffs to U.S. prices was essentially unity, I obtain a downward correction using the measured exit shares, the model’s selection ordering, and the calibration of exiter passthrough. Nevertheless, the scale of the correction is modest, and despite a substantial recent literature on tariff avoidance strategies like transshipment, calibrated cohort passthrough during that episode is close to one.

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A Theoretical Appendix

The theoretical appendix is organized in two parts. The first proves the framework's results from Section 2, which apply across microfoundations. The second establishes the pricing, passthrough, and selection results for the structural model in Section 3. Extensions of the model appear in the supplemental theoretical appendix.

A.1 Appendix to Section 2

A.1.1 Proof of Proposition 1

Proof. By the definitions of the cohort and survivor estimands and $\mathbb{E}_\omega[\sigma_i] = 1 - \theta > 0$,

$$\begin{aligned}\beta_{\text{survivor}} - \beta_{\text{cohort}} &= \frac{\mathbb{E}_\omega[\beta_i \sigma_i]}{1 - \theta} - \mathbb{E}_\omega[\beta_i] \\ &= \frac{\text{Cov}_\omega(\beta_i, \sigma_i)}{1 - \theta}.\end{aligned}\quad \square$$

A.1.2 Proof of Proposition 2

Proof. If $\theta = 0$, then $\sigma_i = 1$ almost surely under the pre-reform weights, so survivor and cohort passthrough coincide and both intervals collapse to β_{survivor} . For $\theta > 0$, splitting the original activity into retained and exiting activity gives

$$\beta_{\text{cohort}} = (1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\text{exiter}}. \quad (\text{A.1})$$

Thus $\beta_{\text{survivor}} - \beta_{\text{cohort}} = \theta(\beta_{\text{survivor}} - \beta_{\text{exiter}})$. By Proposition 1, a nonnegative covariance implies $\beta_{\text{exiter}} \in [\beta_{\min}, \beta_{\text{survivor}}]$, while a nonpositive covariance implies $\beta_{\text{exiter}} \in [\beta_{\text{survivor}}, \beta_{\max}]$. Substituting the endpoints of each interval into equation (A.1) gives the respective bounds. $\square$

A.1.3 Proof of Proposition 3

Proof. If $\theta = 0$, the result is immediate. If $\theta > 0$, the exit-weighted average β_{exiter} inherits the support $[\beta_{\min}, \beta_{\max}]$. Substituting these endpoints into equation (A.1) gives equation (7).

The interval is sharp conditional on survivor passthrough, the exit share, and the support restriction. Fix $b \in [\beta_{\min}, \beta_{\max}]$ and assign passthrough β_{survivor} to retained activity of baseline-weighted mass $1 - \theta$ and passthrough b to exiting activity of mass θ . Binary retention is permitted, so this construction preserves survivor passthrough and the exit share and gives $\beta_{\text{cohort}} = (1 - \theta)\beta_{\text{survivor}} + \theta b$. Varying b over its support spans the entire interval. $\square$

A.1.4 Proof of Proposition 4

Proof. Partitioning exit across margins gives $\theta\beta_{\text{exiter}} = \theta_L\bar{\beta}_L + \theta_H\bar{\beta}_H$. Substituting into equation (A.1) and using $\theta = \theta_L + \theta_H$ yields

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = (\theta_L + \theta_H) \left[\beta_{\text{survivor}} - \frac{\theta_L\bar{\beta}_L + \theta_H\bar{\beta}_H}{\theta_L + \theta_H} \right],$$

which is equation (11). Since $\theta > 0$, the wedge is positive if and only if survivor passthrough exceeds the exit-share-weighted average. Writing $s_L = \theta_L/(\theta_L + \theta_H)$, this condition becomes $s_L(\bar{\beta}_H - \bar{\beta}_L) > \bar{\beta}_H - \beta_{\text{survivor}}$. Because $\bar{\beta}_H > \beta_{\text{survivor}} \geq \bar{\beta}_L$, division by $\bar{\beta}_H - \bar{\beta}_L > 0$ gives the share condition in equation (10). Equality makes the wedge zero; reversing the inequality makes it negative. $\square$

A.1.5 Proof of Corollary 1

Proof. The group-specific interval restrictions imply $\beta_{\text{exiter}} \leq \bar{\beta}_{\text{exiter}}$. Since $\theta = \theta_L + \theta_H > 0$, equation (A.1) gives

$$\theta(\beta_{\text{survivor}} - \bar{\beta}_{\text{exiter}}) \leq \beta_{\text{survivor}} - \beta_{\text{cohort}} \leq \theta(\beta_{\text{survivor}} - \beta_{\text{exiter}}).$$

The left bound is strictly positive when $\beta_{\text{survivor}} > \bar{\beta}_{\text{exiter}}$, giving part (i). The right bound is strictly negative when $\beta_{\text{survivor}} < \beta_{\text{exiter}}$, giving part (ii). $\square$

A.1.6 Proof of Proposition 5

Proof. If $\theta = 0$, survivor and cohort passthrough coincide. Otherwise, the cohort decomposition assigns nonnegative weights θ_L and θ_H to the two exiter averages; any group with zero share may be omitted. Substituting the lower endpoints for both averages therefore gives the lower bound in equation (14), and substituting the upper endpoints gives the upper bound. Under the direction-only restrictions, the lower endpoints for groups L and H are $\beta_{\min}$ and β_{survivor} , respectively, while the upper endpoints are β_{survivor} and $\beta_{\max}$. Substitution and collection of the survivor terms give equation (15). $\square$

A.1.7 Sharpening under Threshold Selection

Corollary 2 (Sharpening under threshold selection). *Suppose $0 < \theta < 1$, exit operates through a single margin, and $\sigma_i \in \{0, 1\}$ is a monotone threshold function of type ξ_i . Let F denote the baseline transaction-value-weighted distribution of ξ , with interval support and continuous density $f > 0$ in*

its interior, and suppose $\beta(\cdot)$ is monotone and continuously differentiable. Vary the threshold while holding F and $\beta(\cdot)$ fixed, and define

$$\beta_{\text{marginal}}(\theta) \equiv \beta_{\text{survivor}}(\theta) - (1 - \theta) \frac{\partial \beta_{\text{survivor}}(\theta)}{\partial \theta}.$$

Then:

(i) If $\text{Cov}_{\omega}(\beta_i, \sigma_i) > 0$, then $\beta_{\text{marginal}} \leq \beta_{\text{survivor}}$ and

$$\beta_{\text{cohort}} \in \left[(1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\min}, (1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\text{marginal}} \right].$$

(ii) If $\text{Cov}_{\omega}(\beta_i, \sigma_i) < 0$, then $\beta_{\text{marginal}} \geq \beta_{\text{survivor}}$ and

$$\beta_{\text{cohort}} \in \left[(1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\text{marginal}}, (1 - \theta)\beta_{\text{survivor}} + \theta\beta_{\max} \right].$$

Each interval is contained in the corresponding one-sided interval of Proposition 2, with strict containment when $\beta(\cdot)$ is nonconstant on the survivor set.

Proof. Without loss of generality, take $\beta(\cdot)$ to be nondecreasing; the decreasing case follows by reversing the type ordering. In case (i), low types exit at a cutoff $\xi^*(\theta)$ satisfying $\theta = F(\xi^*(\theta))$, so $d\xi^*/d\theta = 1/f(\xi^*)$. Writing $M(\theta) = (1 - \theta)\beta_{\text{survivor}}(\theta)$, differentiation gives

$$\begin{aligned} M(\theta) &= \int_{\xi^*(\theta)}^{\infty} \beta(\xi) f(\xi) d\xi, \\ M'(\theta) &= -\beta(\xi^*(\theta)) = -\beta_{\text{survivor}}(\theta) + (1 - \theta) \frac{\partial \beta_{\text{survivor}}(\theta)}{\partial \theta}. \end{aligned}$$

Thus $\beta_{\text{marginal}} = \beta(\xi^*(\theta))$. Since exiters lie below the cutoff and survivors above it, monotonicity implies $\beta_{\text{exiter}} \in [\beta_{\min}, \beta_{\text{marginal}}]$ and $\beta_{\text{marginal}} \leq \beta_{\text{survivor}}$. Substitution into equation (A.1) gives case (i).

In case (ii), high types exit at a cutoff $\bar{\xi}(\theta)$ satisfying $\theta = 1 - F(\bar{\xi}(\theta))$. Now $d\bar{\xi}/d\theta = -1/f(\bar{\xi})$ and $M(\theta) = \int_{-\infty}^{\bar{\xi}(\theta)} \beta(\xi) f(\xi) d\xi$. The same differentiation gives $M'(\theta) = -\beta(\bar{\xi}(\theta))$, so $\beta_{\text{marginal}} = \beta(\bar{\xi}(\theta))$. Monotonicity implies $\beta_{\text{survivor}} \leq \beta_{\text{marginal}}$ and $\beta_{\text{exiter}} \in [\beta_{\text{marginal}}, \beta_{\max}]$, yielding case (ii) by equation (A.1). In either case, the comparison between marginal and survivor passthrough is strict when β is nonconstant on the survivor set, establishing strict containment. $\square$

A.2 Appendix to Section 3

A.2.1 Assumptions about Demand

I impose several regularity conditions on demand:

Assumption 2 (Demand). *The demand for each variety i depends on its own delivered price $\tilde{p}_i$ and on a price aggregator $\mathbf{P}$ summarizing rivals. Fixing $\mathbf{P}$ as parametric at the variety level, the primitives satisfy:*

D1. Downward-sloping own demand and outward shifts in rivals:

$$\frac{\partial D_i}{\partial \tilde{p}_i} < 0, \quad \frac{\partial D_i}{\partial \mathbf{P}} > 0.$$

D2. Smoothness: $D_i(\cdot \mid \mathbf{P}) \in C^3$ in own price.

D3. Elastic demand:

$$\varepsilon_i(\tilde{p}_i \mid \mathbf{P}) \equiv -\frac{\tilde{p}_i}{D_i} \frac{\partial D_i}{\partial \tilde{p}_i} > 1.$$

D4. Interior pricing: For every $m \in [\underline{m}, \overline{m}]$ and channel $r \in \{d, a\}$, the objective

$$(p - m)D((1 + \tau^r)p \mid \mathbf{P})$$

attains an interior maximum at some $p > m$.

D5. Kimball curvature:

$$\kappa_i(\tilde{p}_i \mid \mathbf{P}) \equiv -\frac{\partial \ln \varepsilon_i(\tilde{p}_i \mid \mathbf{P})}{\partial \ln \tilde{p}_i} \leq 0,$$

with $\kappa = 0$ under CES. The restriction also requires:

(a)

$$\frac{\partial \kappa(\tilde{p}_i \mid \mathbf{P})}{\partial \ln \tilde{p}_i} \leq 0;$$

(b) defining

$$\kappa'(\tilde{p}) \equiv \frac{\partial \kappa(\tilde{p})}{\partial \ln \tilde{p}},$$

$$\kappa'(\tilde{p}) \geq -\frac{\varepsilon(\tilde{p})}{\varepsilon(\tilde{p}) - 1} \kappa(\tilde{p})^2,$$

with strict inequality wherever $\kappa(\tilde{p}) < 0$.

Conditions **D1–D4** ensure that the firm's pricing problem is well defined and has an interior solution. Condition **D5** introduces Kimball-type demand, where $\kappa \leq 0$ means that demand elasticity is weakly increasing in price. It also makes the Lerner equation single-crossing and ensures that passthrough is monotone in marginal cost. The empirical literature consistently finds heterogeneous markups and passthrough across firms (Gopinath and Itskhoki 2010; Amiti, Itskhoki, and Konings 2019; De Loecker, Eeckhout, and Unger 2020; Edmond, Midrigan, and Xu 2023); Kimball demand provides a tractable way to generate that heterogeneity.

A.2.2 Proof of Lemma 1

Proof. Fix a channel r , write $t = 1 + \tau^r$, and suppress the fixed aggregator $\mathbf{P}$. By **D4**, an interior optimum exists, and its first-order condition is

$$D(tp) + (p - m)tD'(tp) = 0 \iff \frac{p - m}{p} = \frac{1}{\varepsilon(tp)}.$$

To establish uniqueness, define the Lerner residual $\mathcal{L}(p, m; \tau^r) = (p - m)/p - 1/\varepsilon(tp)$. Under **D5**, $\varepsilon' \geq 0$, so $\mathcal{L}_p = m/p^2 + t\varepsilon'(tp)/\varepsilon(tp)^2 > 0$. The first-order condition therefore has a unique solution.

At the optimum, write $\tilde{p} = tp^*$ and evaluate ε and κ at $\tilde{p}$. Applying the implicit function theorem and using the Lerner condition and $\kappa = -\tilde{p}\varepsilon'/\varepsilon$ gives

$$\frac{\partial p^*}{\partial m} = \frac{\varepsilon}{\varepsilon - 1 - \kappa} > 0, \quad \frac{\partial p^*}{\partial \tau^r} = \frac{\kappa p^*}{t(\varepsilon - 1 - \kappa)} \leq 0.$$

The tariff derivative is zero under CES. Nevertheless, the delivered price rises with the tariff because $\partial \tilde{p}/\partial \tau^r = p^*(\varepsilon - 1)/(\varepsilon - 1 - \kappa) > 0$. It also rises with marginal cost since t is fixed when m changes. Downward-sloping demand therefore implies that quantity falls strictly with both marginal cost and the tariff. $\square$

Lemma 1 supplies the sorting mechanism. Higher-cost firms charge higher prices and sell less. They are therefore closer to the participation cutoff, while lower-cost firms sell more and derive a larger total gain from accessing the lower indirect-channel wedge.

A.2.3 Proof of Proposition 6

Proof. Fix a channel r and write $\tilde{p} = (1 + \tau^r)p^*$. The pricing derivative from Lemma 1 gives

$$\beta = 1 + \frac{1 + \tau^r}{p^*} \frac{\partial p^*}{\partial \tau^r} = \frac{\varepsilon - 1}{\varepsilon - 1 - \kappa},$$

where ε and κ are evaluated at $\tilde{p}$. Since $\varepsilon > 1$ and $\kappa \leq 0$, passthrough lies in $(0, 1]$, equals one under CES, and is strictly below one when $\kappa < 0$.

Using $\partial \varepsilon / \partial \ln \tilde{p} = -\kappa \varepsilon$, differentiation yields

$$\frac{\partial \beta}{\partial \ln \tilde{p}} = \frac{\varepsilon \kappa^2 + (\varepsilon - 1) \kappa'}{(\varepsilon - 1 - \kappa)^2}.$$

By **D5(b)**, this derivative is nonnegative, and positive when $\kappa < 0$. Lemma 1 gives $\partial \tilde{p} / \partial m > 0$, so passthrough has the same monotonicity in marginal cost. $\square$

A.2.4 Proof of Proposition 7

Proof. By **D4** and Lemma 1, optimized variable profit has a unique interior maximizer in each channel. The envelope theorem gives $V_m(m, 1 + \tau^r) = -q^r(m) < 0$. Since $\tau^a < \tau^d$, the lemma also implies $q^a(m) > q^d(m)$, so $\Delta V_m(m) = -q^a(m) + q^d(m) < 0$.

Cutoffs. Continuity and strict monotonicity, together with the fixed-cost restrictions in Assumption 1, give unique cutoffs satisfying

$$V(m_p, 1 + \tau^d) = F_d, \quad \Delta V(m_a) = F_a, \quad \underline{m} < m_a < m_p < \bar{m}.$$

In particular, the restriction on F_d places m_p inside the cost support, and $\Delta V(m_p) < F_a < \Delta V(\underline{m})$ places m_a strictly below m_p and above $\underline{m}$.

Partition. Let $\pi_d(m) = V(m, 1 + \tau^d) - F_d$ and $\pi_a(m) = V(m, 1 + \tau^a) - F_d - F_a$. For $m < m_a$, direct profit is positive and $\Delta V(m) > F_a$, so $\pi_a(m) > \pi_d(m) > 0$ and transshipment is optimal. For $m_a \leq m \leq m_p$, direct profit is nonnegative and $\Delta V(m) \leq F_a$, so direct shipment is optimal. For $m > m_p$, direct profit is negative and $\Delta V(m) < \Delta V(m_p) < F_a$, so $\pi_a(m) < \pi_d(m) < 0$ and exit is optimal. Indifference occurs only at the cutoffs, which have zero mass under continuous G .

Comparative statics. The envelope theorem also gives $\partial V(m, 1 + \tau^r) / \partial \tau^r = (p^* - m)p^* D'(\tilde{p}) < 0$. Thus $\Delta V_{\tau^d} > 0$ and $\Delta V_{\tau^a} < 0$. Applying the implicit function theorem at the corresponding cutoffs yields

$$\begin{aligned} \frac{\partial m_a}{\partial \tau^d} &= -\frac{\Delta V_{\tau^d}}{\Delta V_m} > 0, & \frac{\partial m_a}{\partial \tau^a} &= -\frac{\Delta V_{\tau^a}}{\Delta V_m} < 0, \\ \frac{\partial m_p}{\partial \tau^d} &= -\frac{V_{\tau^d}}{V_m} < 0, & \frac{\partial m_p}{\partial F_d} &= \frac{1}{V_m} < 0. \end{aligned}$$

Here the denominator signs follow from $V_m < 0$ and $\Delta V_m < 0$. □

B Screening Implementation Details

This appendix documents the construction of the transshipment share θ^a and the participation share θ^p . I first describe the data, candidate hubs, and tariff measures, then give the screening and aggregation procedures for each exit margin.

B.1 Data and Notation

I study annual bilateral trade flows using the BACI dataset from CEPII (classification HS2012, V202501 release), covering the years 2012 to 2023. I let $v_{i \rightarrow j, k, t}$ denote the customs value of shipments of product k (at the HS6-digit level) from country i to country j in year t , reported in

thousands of dollars. I keep the data in current dollars when evaluating the viability of transshipment routes, but convert the aggregate exit values to 2017 dollars using the BLS’s import price index (normalized to 1 in 2017) when measuring the value of goods exiting through each channel.

To measure exits from participation, I supplement the BACI data with the US Census Bureau’s detailed import data (HTS10 classification) for China over the period 2011-2023. I use December records, which report both U.S. imports for consumption from China that month and cumulative imports for the calendar year. I use the cumulative yearly value to measure contractions, but the December value to identify categories where trade has ceased by the end of the year. I include 2011 to have a comparison year for the earliest pre-tariff window. I use the product code links provided by the Census Bureau to account for changes in the product classification over the sample, and link the HTS10 classification to the HS2012 classification used for the rest of the data. These links, as well as the product-level data used to measure participation, are described in Appendix B.8.

Given a product k and potential transshipment hub i , I let $\ell_{k,i,t} \equiv v_{\text{CHN} \rightarrow i,k,t}$ and $x_{k,i,t} \equiv v_{i \rightarrow \text{USA},k,t}$ denote the values of the two legs of the potential transshipment route. To evaluate how a hub’s market shares compare between the US and other destinations, I let

$$s_{k,i,t}^{\text{US}} \equiv \frac{v_{i \rightarrow \text{USA},k,t}}{\sum_j v_{j \rightarrow \text{USA},k,t}} \text{ and } s_{k,i,t}^{\text{ROW}} \equiv \frac{\sum_{j \neq \text{USA}} v_{i \rightarrow j,k,t}}{\sum_{j \neq \text{USA}} \sum_m v_{m \rightarrow j,k,t}},$$

where the sums over origins in the denominators range over all countries, not just China and potential transshipment hubs.

I calculate the headline exit shares over the set of 3934 HS6 products that had positive direct trade from China to the US in 2017 and were subject to a positive, completed tariff increase between 2018 and 2019 (defined below). Products that do not satisfy these criteria, due to incomplete data for the tariff increase, are excluded. I denote this product set $\mathcal{K}$, and define baseline (2017) direct trade values at the product level as $b_k \equiv v_{\text{CHN} \rightarrow \text{USA},k,2017}$ and in aggregate as $B \equiv \sum_{k \in \mathcal{K}} b_k$. This denominator is fixed throughout the sample, so that products that do not exit still contribute to the denominator. When I calculate exit shares for placebo products, I normalize by the value of those products’ direct trade in 2017.

B.2 Hub Countries

I start from the list of hub economies provided by CEA (2026), and add economies that, due to their geographical location, host important ports and show strong trade with China and the US, such as Hong Kong and Macao. Table B.1 shows the list of 73 economies considered as potential hubs in my analysis. I consider the same set of potential hubs in each year. This does not mean that all

these economies will be considered as hubs, but rather that they are candidates, and they will be considered hubs only for the goods that satisfy the conditions described in the next subsection.

Table B.1: Candidate Transshipment Hubs (73 Economies)

ARE	ARG	AUT	AZE	BEL	BGD	BGR	BHR	BRA	CAN
CHE	CHL	COL	CRI	CYP	CZE	DEU	DNK	DOM	EGY
ESP	EST	FIN	FRA	GEO	GRC	HKG	HRV	HUN	IDN
IND	IRL	ISR	ITA	JOR	JPN	KAZ	KEN	KHM	KOR
LAO	LKA	LTU	LUX	LVA	MAC	MAR	MEX	MLT	MMR
MYS	NLD	NPL	OMN	PAK	PAN	PER	PHL	POL	PRT
ROU	RUS	S19	SAU	SGP	SVK	SVN	SWE	THA	TUR
UZB	VNM	ZAF							

Notes: Entries are ISO-3 identifiers except S19, which denotes Taiwan (*Other Asia, nes*) in BACI. The implementation retains both BACI country codes for Germany, 276 and 280, so the 73 economies correspond to 74 country codes.

B.3 Tariff Data Sources

Prior to the Global Tariff Database covering China’s direct tariff, I follow Amiti, Redding, and Weinstein (2020) to construct the product-level monthly rate from HTS10 over 2016 and 2017. I average log gross tariff rate by year, transform it back to the tariff rate, and then aggregate to the HS6 product level, using fixed 2017 U.S. import-value shares from China across HTS10 categories within each HS6 product. For the years prior to 2016, I fix the tariff at the 2016 rate.

I measure tariff exposure for each product by looking at the difference in tariff rate for the same product, as long as tariff data are available for both dates in the Global Tariff Database. In particular, I follow Teti (2025)’s methodology to calculate a yearly average for each product from the dated schedules, by averaging the log gross tariff over days and then transforming back to the tariff rate. To calculate the completed tariff increase, I take the difference between the last recorded tariff rate in 2019 and the tariff rate that was imposed on Jan 1 2018, before the Section 301 tariffs were increased. The product-level tariff rate in the Global Tariff Database are at the HS2017 level; I convert this to HS2012 used in BACI using the concordance table from the UN Statistics Division. In cases where multiple HS2017 codes are converted to a single HS2012 product, I use the 2017 US Import Value from China in each product, according to the US Census data, as the weight to aggregate. For the product to be included in $\mathcal{K}$, the tariff comparison must be available, and the tariff must have been increased. For the year-by-year changes used in the regressions, I calculate the log gross tariff change compared to the pre-tariff reform schedule,

using the same method for matching and weighting. For the incidence calculation in Section 4.2, I use the change in the tariff rate from the completed tariff increase.

In the screening steps, I use the country-specific tariff for each hub in each product, as reported in the Global Tariff Database, and denote them $\tau_{k,i,t}^{\text{hub}}$, to distinguish it from the average hub tariff for a product, denoted $\tau_{k,t}^h$, used in the product-level regressions. The average hub tariff is $\tau_{k,t}^h = \sum_{i \in \mathcal{H}} w_{k,i}^h \tau_{k,i,t}^{\text{hub}}$, where $w_{k,i}^h$ is the US import share of product k from hub i out of all candidate hubs, at the year 2017.¹⁹ Given that the hub tariff data starts at 2018, for the years between 2012 and 2017, I fix the average tariff at the 2018 value. Then I calculate the gap between the direct and hub tariffs at the product level, as $\Delta_{k,t} = \log \left(\frac{1 + \tau_{k,t}^d}{1 + \tau_{k,t}^h} \right)$, and use the individual hub tariffs in the route-level comparison, applying the same timing conventions as above.

B.4 Trend Estimation

I estimate trends separately for each China-market and hub-market share, for each market share outside the US, and for each trade leg value. I use data from the 6 years prior to the tariff increase, from 2012 to 2017, and do not update the trend using any post-tariff observations. Prior to 2018, the before and after comparisons are calculated using the same data as the trend estimation; from 2018 onward, it is calculated using the forecasted value.

I use a linear specification, estimating the slope of the logit of shares and the log of trade value. Then I convert the predicted value back to the share and the trade value respectively. For the trade value, I take the log after replacing the value with $\max\{v, 1\}$, where the unit of trade value in the BACI data is 1000 USD.²⁰

A minimum of 3 observations in the pre-period is required to estimate the trend. For the market share, a minimum of 2 observations between (0, 1) strictly is required to estimate the trend in the logit; for observations with less than 2 non-boundary observations, I calculate the arithmetic mean of the pre-period observed value. For trade value, if less than 3 observations are available for the trend, I use the pre-period observed value's arithmetic mean. For any year where the value is not observed in the data, I do not assume that the trade value is 0. In the calculation of route-level outcomes, if the market share information is not available, I do not consider it a qualifying route. If the trade value forecast for a leg is not available, I set the predicted value for that leg to be 0, but it must still have observed trade for both legs and pass the other screens in order to be considered a qualifying route.

19. For the calculation of average hub tariff, I use 0 to fill in missing cells. If a product is not sold by any of the candidate hubs to the US, I use an unweighted average tariff across hubs.

20. For the logit transformation, I take the log after replacing both s and $1 - s$ with $\max\{s, 10^{-12}\}$ and $\max\{1 - s, 10^{-12}\}$.

B.5 Screening Criteria

I construct a measure of transshipped trade that requires both tariff savings along a transshipment route and a corresponding shift in the pattern of trade from China through a potential hub country to the United States. In the low and high screens, a potential transshipment route is identified for product k through hub country i in year t if five conditions hold:

1. The product is in the tariff-exposed cohort $\mathcal{K}$, and the direct tariff is above its pre-reform level in the year for which transshipped trade is measured. Specifically, I use a day-weighted harmonized tariff from the tariffs in Appendix B.3, and therefore impose this condition in each year.
2. The tariffs on this route are lower than the direct tariffs, $\tau_{k,i,t}^{\text{hub}} < \tau_{k,t}^d$. I use the tariffs from the same year in 2018 and 2019 and set them equal to the 2019 tariffs afterwards. I impose the condition on direct tariffs in each year.²¹
3. The shift in trade patterns condition in the United States holds, $s_{k,\text{CHN},t}^{\text{US}} \leq \widehat{s}_{k,\text{CHN},t}^{\text{US}} - \delta$ and $s_{k,i,t}^{\text{US}} \geq \widehat{s}_{k,i,t}^{\text{US}} + \delta$, where $\delta = 10^{-6}$ is a numerical tolerance level in share terms.
4. For the countries outside of the United States, the hub country's deviation from its pre-trend market share is no larger than China's deviation, $s_{k,i,t}^{\text{ROW}} - \widehat{s}_{k,i,t}^{\text{ROW}} \leq s_{k,\text{CHN},t}^{\text{ROW}} - \widehat{s}_{k,\text{CHN},t}^{\text{ROW}}$. I use this condition even if it is negative to ensure that the hub country's competitiveness is not improving outside of the US for other reasons.
5. The level of exports from the hub country and the level of imports to the hub country from China both rise relative to their pre-trend values, and the level of the smaller of the two legs of the transshipment route is above some minimum level: $\ell_{k,i,t} > \widehat{\ell}_{k,i,t}$, $x_{k,i,t} > \widehat{x}_{k,i,t}$, and $\min\{\ell_{k,i,t}, x_{k,i,t}\} \geq 200$, where the final value is in units of 1,000s of current dollars as defined in the BACI data.

In addition to the above conditions, I require that a transshipment route is persistent. Specifically, I require that conditions 3 through 5 hold for a route in an adjacent year. Therefore, if I identify a transshipment route in 2019, for example, it must also satisfy these conditions in 2018 or 2020 for it to be considered as a route in 2019. I impose this requirement within my sample periods of 2012-2017 and 2018-2023, so that conditions 3 through 5 holding for a route in 2017 and

21. For the post-2019 comparison, where a 2019 route tariff is unavailable, I use its 2018 value. I drop product-country-year observations for which I do not observe both the direct and hub tariffs. In the main analysis, I use a tolerance level of 10^{-12} when comparing the hub and direct tariffs so that I am imposing the condition strictly. In placebos that I discuss below, I do not impose the condition strictly so that I also allow the hub tariffs to equal the direct tariffs.

2018 will not qualify the route as a transshipment route in either 2017 or 2018.²² This adjacent year need not satisfy the tariff conditions. Instead, I impose the tariff conditions after identifying routes that satisfy the trade pattern conditions persistently. Supplemental Table S1 reports my estimates of the transshipment response to tariffs under alternative tariff-eligibility rules.

B.6 Transshipment Values and Aggregation

The low and high screen differ in the amount of trade that is attributed to each qualifying transshipment route. The low screen attributes the lower of the two increases in trade along the two legs of the route, so that only increases in trade along the route are considered. The high screen, instead, attributes the lower of the two levels of trade along the two legs of the route, so that pre-existing trade along the route is also attributed. Let $\mathcal{R}_t$ denote the set of transshipment routes that pass the trade pattern, persistence, and hub tariff conditions. Then, I define the total value of transshipped trade for product k in year t under the low and high screen as

$$A_{k,t}^L = \sum_{i:(k,i) \in \mathcal{R}_t} \min\{\ell_{k,i,t} - \widehat{\ell}_{k,i,t}, x_{k,i,t} - \widehat{x}_{k,i,t}\} \text{ and } A_{k,t}^H = \sum_{i:(k,i) \in \mathcal{R}_t} \min\{\ell_{k,i,t}, x_{k,i,t}\}, \quad (\text{A.2})$$

in units of current dollars and 1,000s, and equal to zero for products that do not have any identified routes.

I cap the value of transshipped trade relative to its baseline both in nominal terms, so that $A_{k,t}^s$ is no larger than the value of direct trade from China to the US in 2017, b_k , and in real terms relative to the import price index, which I normalize to be equal to one in 2017, P_t .²³ I then define the share of a product that is transshipped under screen $s \in \{L, H\}$ as

$$\check{\theta}_{k,t}^{a,s} = \min \left\{ \frac{\min\{A_{k,t}^s, b_k\}}{P_t b_k}, 1 \right\}. \quad (\text{A.3})$$

Although I require the trade patterns along a route to be persistent, the share of a product that is identified as transshipped does not need to be persistent because the tariffs may not allow a route to qualify in the adjacent year. I therefore remove years in which I identify a positive product share after 2017 if I do not identify a positive product share in either adjacent year, which

22. At the boundaries of my sample, I require 2012 to have an adjacent year in 2013 and 2023 to have an adjacent year in 2022.

23. To get the external comparison at the hub level, I proportionally allocate the total value of transshipped trade from the low screen to the hub countries based on their value in the low screen and then allocate the additional value from the high screen proportionally based on the difference between the two screens at the hub level. This ensures that I am using the same total value of transshipped trade for a product and that the value of transshipped trade for a hub country is always greater under the high screen.

I do for both the low and high screens. I impose this condition after imposing the route-level tariff conditions and caps but before imposing the condition on direct tariffs in each year.

After imposing the annual direct-tariff requirement, I restrict later-year estimates to products detected in 2018 or 2019. That is, if a product was detected to transship in 2018 or 2019 (had positive measured transshipment in either year), then I measure its transshipment in the subsequent years 2020 to 2023 based solely on the behavior of this group of products. In other words, the measured transshipment in the subsequent years can still route through different hubs and can rise, fall, or return to zero, but it is only measured for the group of products that were initially detected.

I denote the resulting product-level share after imposing all of these restrictions as $\theta_{k,t}^{a,s}$. The aggregate share is simply the average of these product-level shares, weighted by their direct trade values in 2017: $\theta_t^{a,s} = \sum_{k \in \mathcal{K}} \frac{b_k}{B} \theta_{k,t}^{a,s}$, where $B = \sum_{k \in \mathcal{K}} b_k$ includes all products in the original cohort, including those for which no transshipment was detected.

B.7 Pre-Tariff and Never-Treated Product Comparisons

To consider the extent to which trade patterns could satisfy the screens for reasons unrelated to the tariff increases, I also perform the following comparisons.

First, I apply the share, flow, and persistence conditions, but to the same original product cohort over the years 2012 to 2017, prior to the tariff increases. For this series, the pre-trends are fixed as described above and the value is normalized by the same cohort's 2017 direct trade value. This comparison does not require a product to experience a tariff increase prior to 2018, but it still requires that its tariffs satisfy the comparison condition, excluding hubs that face higher tariffs than China.²⁴

Second, I apply the conditions to products that did not experience any changes in their direct tariffs between the pre-tariff year and the average of the two reform years.²⁵ There are 177 products that satisfy this condition, of which 143 had positive direct trade in 2017 and appear in the reporting panel. For this group, I use their own pre-trend history when estimating transshipment prior to the tariff increases and I normalize their transshipment by their own 2017 trade base. I maintain the requirement that the trade patterns satisfy the conditions, including the requirement that the routes remain persistent over time, and I allow the tariffs on the routes through the hubs to be the same as the direct tariff from China.²⁶

24. Because I do not have tariff data for the hubs prior to 2018, I compare China's 2017 tariff rate to the lower of the hub's 2018 and 2019 tariff rates. I allow for the tariffs to be equal in this case. The nominal and real cap values are the same as in the main calculation.

25. I identify products to include in this group as those for which the average of their log gross tariffs in 2018 and 2019 is the same as their log gross tariff in 2017, up to a tolerance of 10^{-12} .

26. In this case, I use the 2018 and 2019 tariffs for 2018 and 2019, respectively, and I use the 2019 tariffs (or 2018

This comparison therefore allows for detections to occur after 2019 and measures the extent to which trade is reallocated over the years 2018 to 2023 for reasons not related to the tariff increases. At their highest in 2022, this comparison yields transshipment shares of 0.82% and 1.36% under the low and high screens, respectively. These are the placebo shares that I report without restriction in Section 4.3. When comparing these shares to the main panel, however, I restrict the later years to the set of products that were detected in 2018 to 2019 (Supplemental Figure S1), as I do for the main calculations. Regardless, these placebo shares are not used to determine the values of the screens.

B.8 Participation Exit Proxy

I consider a category to have exited participation if it ceases participation in the US market. The participation margin is measured using either disappearance or severe contractions in detailed categories of the HTS10 classification, excluding products in HS6 codes with substantial measured transshipment. Some categories within a single HS6 product may be excluded, leaving a non-zero baseline level of trade within that product. Participation exit is initially identified in 2018–2019 and re-measured in each year through 2023.

Data pre-processing. I use Census imports for consumption data from China. For December of each year, these data report both the monthly value of trade and the cumulative value over the calendar year. I use the monthly December value to determine presence in December of each year. I deflate the annual value by the import price index to measure real trade in 2017 dollars. I use data from 2012–2023, but 2011 is available as a reference year for the first pre-tariff comparison.

I avoid mistaking HTS10 codes replaced with new codes as exits by merging categories using the historical import code concordance from the Census and USITC. I merge codes that are linked through all classification changes between 2012 and 2019, and sum their trade in each year. I refer to these codes as harmonized HTS10 categories (or simply harmonized categories), indexed by ℓ . I map each 2017 harmonized category to an HS2012 product (which may be more than one product if the category was split across multiple HS6 codes by 2017), which I use when summing baseline trade across categories within products. In subsequent years, I again merge and split categories to account for changes up to the year of measurement, and sum both current and historical trade for each harmonized category. I also use the codes in each harmonized category to look for continued presence of the category in the most recent year in data, in case the original codes have a successor that is valid in the year of measurement.

where 2019 data are not available) for subsequent years. Additionally, because these products are not in the treated group, I do not apply the additional product-level check for isolated single-year post-2017 detections.

Initial identification. Let $v_{\ell,t}$ be real annual trade for each category ℓ in year t , and let $\widehat{v}_{\ell,t}$ be the predicted level of trade from a log-linear trend estimated using data from 2012 to 2017. I calculate the ratio of actual trade to the minimum of predicted and actual 2017 trade as $r_{\ell,t} = v_{\ell,t} / \min\{\widehat{v}_{\ell,t}, v_{\ell,2017}\}$. I interpret a low value of this ratio as implying both a substantial decline from the predicted value and from the 2017 level. The ratio requires a decline from the baseline, so that categories with slower-than-predicted growth but no decline do not enter the set of exiting categories.²⁷

A category is initially included if it either had no direct trade in December 2019 or if it is below a threshold ϵ_p^* in both 2018 and 2019, $\{r_{\ell,2018} < \epsilon_p^*, r_{\ell,2019} < \epsilon_p^*\}$. I set $\epsilon_p^* = 0.273$, so that a product must contract to less than 27.3 percent of both its predicted value and its 2017 value to be included through severe contraction. For disappearance, I require that the category had no direct imports in December of 2019, but do not require that it was absent in prior months in 2019 or that it is permanently absent in subsequent years. I do not impose any requirement that the category's exports to other destinations have declined, as a category may exit the US market without exiting other markets.

To determine ϵ_p^* , I consider trade prior to the tariffs. For each two-year window between 2012 and 2017, I fit a log-linear trend to the remaining four years, and consider categories with actual trade below a fraction of predicted trade in both years of the window and that are present in December of the second year. For example, I fit trends using trade from 2012–2015 to compare 2016 to 2017. I choose the largest fraction on a grid from 0 to 1 in increments of 0.001 such that the sum of trade for flagged categories is at most 1 percent of the 2017 value of trade for the tariff-exposed cohort in each comparison. I value each category by multiplying its fraction of 2017 Census trade within its HS6 product by the total value of 2017 BACI trade for the product.

This procedure gives a threshold of $\epsilon_p^* = 0.273$. I then use this threshold when also requiring that trade be below a fraction of actual 2017 trade. The additional requirement that trade decline relative to 2017 can only reduce the number of products flagged by the procedure. The one percent criterion is imposed only for the severe contraction component, using the comparison to forecasted trade, for each two-year comparison. It is not imposed for disappearance or for the combined measure of annual participation.

I exclude categories in products with a high level of measured transshipment in both 2018 and 2019. I calibrate the level of transshipment used to exclude categories by refitting trends for each of the five two-year windows from the pre-tariff period and constructing the transshipment shares in the two years of each window. I choose the smallest value such that products with transshipment at or above that level in both years account for at most 1 percent of the value

27. Categories with missing years are set to zero trade. Trends are estimated by OLS using the logarithm of the maximum of real trade and 1,000 2017 USD. Predictions are set to a minimum of 1,000 2017 USD.

of 2017 direct trade for the calibration cohort in each two-year window. This procedure gives thresholds of 11.24 percent for the low screen and 26.78 percent for the high screen.²⁸

I use the corresponding low/high transshipment exclusions for my low/high participation measures. However, unlike the transshipment measures, these can initially co-exist for a given product as long as the detected transshipment is below the exclusionary level. I use the same criteria for what counts as a disappearance/contraction. For later years I impose the same stricter exclusion criteria as with transshipment.

As with transshipment, I only look for disappearances and contractions in 2018 and 2019 among the initial set of categories. In 2018 and 2019, an initially selected category contributes only if it has no trade that December or satisfies $r_{\ell,t} < 0.273$ in that year. Starting in 2020, I only consider these categories and any future category must have no trade in December of that year or $r_{\ell,t} < 0.273$ to count that year. If a category had disappeared or contracted in 2019 and then re-appeared it does not automatically count in future years; it must again meet these criteria. As with transshipment, if classifications change such that a non-counting and counting category are combined into a new one, I re-evaluate trade for the new combined group, but only include the baseline value of the initially selected activity.

Additionally in all post-tariff years, for a product in the tariff-exposed cohort to count toward the measure, it must still have a direct applied tariff rate that is higher than its pre-reform level. Also, starting in 2020 I impose the same zero currently-detected transshipment requirement at the parent HS6 product level as with my corresponding high or low transshipment screens. "Currently-detected" here means that I use current-year detections to drop products before imposing the restriction that I only include products first detected in 2018 or 2019 in my reported transshipment series. Thus, I allow new transshipment detections to exclude products from my participation measures even if those new detections are not themselves included in my reported transshipment series. As a result my two margins are disjoint at the HS6 product level for years after 2019.

To assess the frequency of categories meeting my criteria in the pre-tariff period I consider each consecutive two year period from 2012-2017, holding out those two years and using the remaining four years to construct my trends. I compare trade in the two year period not to 2017, but rather to the year just before this two year period (e.g. for 2012-2013 I compare to 2011). In

28. I calibrate the screens on the set of 4107 treated products in the bilateral panel, setting observations of missing products and product-hub combinations to zero. A product is excluded if it meets the trade pattern and flow screens from Appendix B.5 in both years of each window and is confirmed using the adjacent pre-tariff year. I use the comparison of direct and hub tariffs from 2018 in the first year of each window, and 2019 in the second, so that the eligibility of routes is identical to the reform years. Transshipment route values are summed across routes within each product, deflated by the import price index, and truncated to the value of 2017 real trade within each product. I use these held-out comparisons separately from the placebo transshipment exercise in Appendix B.7, which allows any year to be used.

order to do this for the 2012-2013 period I combine categories that were distinct in 2011 due to a January 2012 classification code change. I include a category in a particular two year period if it had no trade in December of the second year or meets my contraction criteria in both years. However, to contribute to the measure in a particular year it must have no trade in December of that year or meet my contraction criteria in that year, and it cannot contribute in any years before its first observed year of trade. I exclude any categories that meet my corresponding held-out transshipment criteria using the same fixed low and high levels as for the post-tariff period. If a category is flagged in two adjacent two year windows in a particular year I only count its baseline value in one of these windows.

Note that these annual pre-period shares are not intended to be interpreted as the fraction of categories that exited for any reason, but rather the fraction that my methodology would flag in a given year. I include my comparison set of products with stable tariffs in these annual regressions and apply the same rules to them, except that they do not need to have had their tariffs raised.

Let $c_{\ell k 2017}$ be the Census baseline value of the harmonized category ℓ in HS6 product k and let $I_{\ell k t}^{p,s}$ be an indicator for whether this category meets the participation and product level exclusion criteria in year t under screen $s \in \{L, H\}$. Then the product level participation share under screen s is $\theta_{kt}^{p,s} = \frac{\sum_{\ell} c_{\ell k 2017} I_{\ell k t}^{p,s}}{\sum_{\ell} c_{\ell k 2017}}$ and the aggregate share is $\theta_t^{p,s} = \sum_{k \in \mathcal{K}} \frac{b_k}{B} \theta_{kt}^{p,s}$. Note that Census gives us the fraction of baseline trade in a product that meets my criteria, but BACI gives us the baseline value of that product, b_k . For instance, if 20% of a particular product's trade meets my criteria and it is 0.5% of baseline trade, then it contributes 0.001 to the aggregate measure. The denominator B is the same as in my transshipment measure; it is the original value of trade subject to the tariffs. Census only provides composition data for 99.93% of this base; products without composition data are still in the denominator but have no measured participation. To get my incidence measures I weight by the exposure-adjusted weights from Section 4.2 instead of trade values.

C Supporting Empirical Results

This appendix reports summary statistics and supporting results for the empirical analysis in Section 5.

C.1 Summary Statistics

Table C.1: Summary Statistics

Variable	Pre		Post		Post-Pre	N
	Mean	SD	Mean	SD		
$\Delta_{k,t}$	0.006	0.027	0.106	0.096	0.100	4196
Capital	0.004	0.015	0.143	0.079	0.140	607
Consumption	0.008	0.031	0.061	0.083	0.054	920
Intermediate	0.005	0.024	0.172	0.079	0.167	2669
Transshipment share, low screen	0.000	0.006	0.029	0.089	0.028	4196
Capital	0.000	0.005	0.049	0.113	0.049	607
Consumption	0.000	0.004	0.009	0.037	0.009	920
Intermediate	0.001	0.009	0.056	0.129	0.055	2669
Transshipment share, high screen	0.003	0.021	0.062	0.162	0.059	4196
Capital	0.002	0.021	0.119	0.213	0.117	607
Consumption	0.003	0.012	0.018	0.064	0.015	920
Intermediate	0.004	0.032	0.117	0.224	0.113	2669
Participation share, low screen	0.003	0.042	0.005	0.058	0.002	4196
Capital	0.004	0.050	0.003	0.041	-0.001	607
Consumption	0.001	0.019	0.001	0.017	-0.000	920
Intermediate	0.007	0.065	0.015	0.103	0.007	2669
Participation share, high screen	0.003	0.042	0.005	0.058	0.002	4196
Capital	0.004	0.050	0.002	0.039	-0.001	607
Consumption	0.001	0.019	0.001	0.017	-0.000	920
Intermediate	0.007	0.065	0.015	0.103	0.008	2669
Any transshipment, low screen	0.066	0.248	0.329	0.471	0.263	4196
Any transshipment, high screen	0.066	0.248	0.329	0.471	0.263	4196
Consumption share (2017)	0.419	0.413	0.419	0.413	-0.000	4196
Capital share (2017)	0.300	0.372	0.301	0.372	0.001	4196
Intermediate share (2017)	0.281	0.421	0.280	0.421	-0.001	4196

Notes: Means and standard deviations are weighted by 2017 China-to-U.S. import exposure. Pre and post refer to 2012–2017 and 2018–2023. Pre-period participation shares use the placebo measures. Indented rows classify HS6 products by dominant 2017 end use. N is the number of products observed in both periods.

C.2 Hub Tariffs and Transshipment

Table C.2: Hub Tariffs and Transshipment

	Transshipment share	
	Low screen	High screen
	(1)	(2)
$\text{Post}_t \times \bar{\tau}_k^h$	-0.002 (0.002)	-0.007** (0.004)
Observations	50,568	50,568
HS6 & Year×Use FE	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Each column regresses the annual transshipment share on $\text{Post}_t \times \bar{\tau}_k^h$ using OLS. Observations are HS6-by-year-by-end-use cells in the tariff-exposed cohort over 2012–2023. $\bar{\tau}_k^h$ is the 2018 U.S. tariff on imports from candidate hubs, averaged using their 2017 shares of U.S. imports of the product and standardized within the sample. $\text{Post}_t = 1$ from 2018 onward. Regressions include HS6 and year-by-end-use fixed effects and use 2017 import-value weights allocated by end-use shares. HS6-clustered standard errors are in parentheses. Supplemental Figure S6 reports the corresponding year-specific estimates.

C.3 Selection Across Exit Margins

Table C.3: Pre-tariff Differences across Exit Margins: Alternative Specifications

	Pre-tariff sales (PPML)		China's pre-tariff U.S. import share (standard deviations)	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
Baseline specification	0.327*** (0.101)	0.326*** (0.101)	0.223*** (0.052)	0.218*** (0.051)
Add pre-tariff route-count controls	0.217** (0.084)	0.212** (0.086)	0.170*** (0.051)	0.164*** (0.050)
Use two-leg route-count fixed effects	0.269*** (0.089)	0.266*** (0.089)	0.160*** (0.051)	0.152*** (0.050)
Also control for pre-tariff hub identities	0.211*** (0.081)	0.209** (0.082)	0.154*** (0.051)	0.147*** (0.050)
Same specification, trade-value weights	0.166* (0.093)	0.163* (0.094)	0.041 (0.066)	0.042 (0.066)
Products	3,934	3,934	3,934	3,934

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Each row reports transshipment-only minus participation-only coefficient differences from separate regressions of 2012–2016 average sales (PPML) and standardized import shares (OLS). All rows retain the controls and HS4 fixed effects in Table 2. The route controls count hubs with both China-to-hub and hub-to-U.S. trade in 2017, with only one of these legs, and with exports to non-U.S. destinations. The third row replaces the two-leg count controls with count fixed effects; the fourth adds indicators for the identities of the two-leg hubs. Rows 1–4 use equal product weights; row 5 uses 2017 trade-value weights. HS4-clustered standard errors are in parentheses.

Table C.4: Pre-tariff Characteristics of Participation Exiters within HS6 Products

	Equal category weights		2017 trade-value weights	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
2017 log unit value	0.278*** (0.037)	0.274*** (0.037)	1.588 (1.101)	1.567 (1.116)
2017 log quantity	-2.822*** (0.082)	-2.819*** (0.082)	-2.725*** (0.801)	-2.701*** (0.813)
2017 log sales	-2.544*** (0.073)	-2.545*** (0.073)	-1.137*** (0.348)	-1.134*** (0.351)
Harmonized HTS10 categories	10,682	10,682	10,682	10,682
HS6 products	3,473	3,473	3,473	3,473

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Observations are harmonized HTS10 categories in Census data. Each cell regresses the indicated 2017 characteristic on a subsequent participation-exit indicator, with HS6-by-physical-unit fixed effects. The comparison group comprises categories not flagged for participation exit. Low/high identify the transshipment screen used in the participation exclusions. Equal category weights assign one to every observation; trade-value weights use baseline category values mapped to the original cohort. Parentheses report HS6-clustered standard errors.

Table C.5: Pre-tariff Unit Values and Subsequent Transshipment

	Transshipment share				Any transshipment			
	Low screen		High screen		Low screen		High screen	
	Diff.	Non-Diff.	Diff.	Non-Diff.	Diff.	Non-Diff.	Diff.	Non-Diff.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Pre-tariff Chinese unit value</i>								
Pre-Tariff Measure	−0.001 (0.007)	−0.005 (0.007)	0.005 (0.011)	−0.015 (0.012)	0.016 (0.036)	−0.018 (0.058)	0.016 (0.036)	−0.018 (0.058)
Observations	2,708	1,195	2,708	1,195	2,708	1,195	2,708	1,195
<i>Panel B. Pre-tariff Chinese unit-value premium relative to the world</i>								
Pre-Tariff Measure	0.001 (0.015)	0.023** (0.011)	0.007 (0.022)	0.037** (0.019)	0.052 (0.061)	0.096* (0.054)	0.052 (0.061)	0.096* (0.054)
Observations	2,862	1,277	2,862	1,277	2,862	1,277	2,862	1,277
HS4 & End-Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each column reports an OLS coefficient using one observation per HS6-by-end-use cell. Outcomes are the average transshipment share over 2018–2023 (columns 1–4) or an indicator for detection in any of those years (columns 5–8). Unit-value measures use 2012–2017 trade. Panel A uses log Chinese unit values standardized within HS4; Panel B uses the standardized log ratio of Chinese to all-origin U.S. import unit values. Regressions are estimated separately for differentiated (‘Diff.’) and non-differentiated (‘Non-Diff.’) products under the Rauch (1999) classification. All specifications control for post-period tariff exposure, pre-tariff China import share, China-to-hub routing intensity, bulkiness, origin concentration, and an electronics indicator; include HS4 and end-use fixed effects; and use 2017 China-to-U.S. import-value weights allocated by end-use shares. Parentheses report HS6-clustered standard errors.

C.4 Exit Shares and Subsequent Direct Trade

Table C.6: Exit Shares and Deviations from Pre-tariff Price and Sales Trends

	Log FOB unit value		Log import value	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
Transshipment share (+10 pp)	−0.073** (0.037)	−0.042** (0.019)	−0.048* (0.025)	−0.032* (0.016)
Participation share (+10 pp)	0.020 (0.022)	0.020 (0.022)	−0.152*** (0.037)	−0.153*** (0.038)
Difference between coefficients	−0.093** (0.039)	−0.063** (0.027)	0.105** (0.045)	0.121*** (0.041)
Observations	51,455	51,455	51,455	51,455
HS6 products	4,081	4,081	4,081	4,081

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: OLS regressions use annual HS6-by-end-use observations with positive China-to-U.S. import values and quantities over 2012–2023. Outcomes are observed logs minus predictions from product-specific 2012–2017 log-linear trends. Unit value is import value divided by reported quantity. Both annual exit shares enter jointly; coefficients correspond to a 10-percentage-point increase in the indicated share. Low/high identify the transshipment screen and corresponding participation exclusions. All columns use the same observations, HS6 and year-by-end-use fixed effects, and 2017 import-value weights allocated by BEC shares. The reported difference is the transshipment coefficient minus the participation coefficient. Parentheses report HS6-clustered standard errors.

D Passthrough Robustness

The selection correction to passthrough measurement is solely concerned with differences in passthrough between activity that remains in the observed tax base and activity that exits. However, it could also be that customs unit values change due to changes in the composition of traded goods. If trade shifts toward cheaper goods, then measured prices could fall even without changes in individual goods' prices, which may offset the overstatement of cohort passthrough caused by selection documented in the main text.²⁹ This part of the appendix decomposes observed changes in unit values into changes in the unit values of continuing HTS10 product categories and changes in the mix of categories traded. I compare the latter component with the calibrated selection correction.

To do the decomposition, I use the detailed HTS10 data from the Census. I compare 2017 with 2019, harmonizing HTS10 categories across classification changes. For categories observed in both years, I distinguish changes in unit values from changes in their share of the HS6 product's total quantity, accounting separately for categories that enter or leave direct trade between the two years.³⁰ To add quantities across categories, I require them to be measured in the same physical unit and available for all recorded trade in both years. The resulting sample contains 3,110 products and covers 57.6 percent of the original cohort's 2017 direct-trade value.³¹

Table D.1: Within-category and Composition Changes in Unit Values, 2017–2019

Weighting	Total	Within	Mix	Entry	Exit
Trade value	-2.636	-1.038	-1.550	-0.043	-0.005
Tariff exposure	-3.170	-1.369	-1.740	-0.053	-0.009

Notes: Entries are log changes multiplied by 100. Products require complete value coverage and a common reported physical unit across constituent detailed categories. Within, mix, entry, and exit sum to the total. Trade-value weights use 2017 cohort values; exposure weights also incorporate the completed-package tariff increase.

Table D.1 shows that composition accounts for a substantial part of the observed decline in unit values. Under trade-value weighting, unit values fell by 2.64 percentage points, of which 1.04 came from within-category changes and 1.60 from composition. Most of the composition component reflects reweighting among continuing categories rather than entry or exit. Under

29. Appendix Table C.6 reports the relationship of observed unit values and sales with the two exit shares.

30. For categories traded in both years, I measure the contribution of changes in their unit values while holding quantity shares fixed at the average of the two years. I separately measure the contribution of changes in quantity shares, using each category's average unit value minus the HS6 product's 2017 unit value. Entry and exit are also evaluated relative to that 2017 unit value. The four contributions sum exactly to the change in the HS6 unit value. I rescale them proportionately so they sum to its log change.

31. Supplemental Table S16 reports coverage under alternative quantity-data requirements.

exposure weighting, composition accounts for 1.80 of the 3.17 percentage point decline.³²

Since prices decline through a change in composition, that negative component could offset the overstatement of cohort passthrough caused by selection. I examine this possibility by adding a composition term to the survivor–cohort comparison. Let $\tilde{\beta}_{\text{cohort}}$ denote the passthrough estimate before correcting for either selection or composition, and let $\bar{c}$ denote the change in unit values due to changes in the mix of goods traded, expressed per unit of the log tariff increase. Here, β_{survivor} denotes price-only survivor passthrough. Moreover, let $\lambda \in [0, 1]$ denote the fraction of the change in prices due to composition that affects estimated passthrough. This is an unknown parameter and is not necessarily equal to one since controls may attenuate the effect of composition on passthrough. Then the corrected passthrough is

$$\beta_{\text{cohort}} = \tilde{\beta}_{\text{cohort}} - (\theta^a (\beta_{\text{survivor}} - \bar{\beta}_a) - \theta^p (\bar{\beta}_p - \beta_{\text{survivor}}) + \lambda \bar{c})$$

I evaluate this correction over 2017–2019 using the 3,109 products with positive direct tariff increases in the decomposition sample. At $\beta_{\text{survivor}} = \bar{\beta}_p = 1$ and $\bar{\beta}_a = 0.525$, the participation term is zero, while the avoidance term $\theta^a(1 - 0.525)$ equals 1.33 and 3.19 percentage points under the low and high screens, respectively. For composition, I divide the average unit-value change attributable to changes in the mix of goods traded by the average log tariff increase, weighting both averages by the 2017 value of activity remaining in direct trade. This gives composition terms of -12.29 and -12.36 percentage points, respectively, if $\lambda = 1$. Consequently, the selection correction outweighs composition if less than 10.8 percent of the measured composition effect enters the passthrough coefficient under the low screen, or less than 25.8 percent under the high screen.

The preceding comparison and the main application both use $\bar{\beta}_a = 0.525$ as an externally anchored calibration of average counterfactual direct-channel passthrough among avoidance ex-its. That calibration comes from Amiti, Itskhoki, and Konings (2019), which find nearly complete own-cost passthrough among small firms and substantially lower passthrough among large firms. The model in this paper motivates calibrating the passthrough of transshippers using their estimates for large firms; larger firms are more likely to transship in the model. Moreover, there is a clear link between their estimated passthrough coefficients and the model passthrough: holding $\mathbf{P}$ fixed, the same curvature factor in equation (18) governs own-cost and tariff passthrough in the model. Their Table 2 reports a large-firm own-cost coefficient of 0.478 in column 2. Column 6, which defines large firms as the top 20 percent by domestic sales within an industry, implies a coefficient of $0.802 - 0.235 = 0.567$. The midpoint of the rounded values is $(0.48 + 0.57)/2 = 0.525$,

32. Supplemental Table S17 regresses the unit-value components on tariff changes and their interactions with the two exit shares.

which is what I use as a baseline.³³

While the baseline calibration is sensible, it is also an unestimated parameter. Consequently, Table D.2 reports sensitivity to alternative passthrough calibrations for both transhipper passthrough and survivor passthrough. Panel A recalibrates transhipper passthrough for the two Amiti, Itskhoki, and Konings (2019) coefficients, and tests two alternative avoidance passthrough estimates of 0.9 and 1.0, respectively. As avoidance exiters become more similar to survivors, the correction shrinks. At $\bar{\beta}_a = 0.9$, survivor passthrough exceeds cohort passthrough by only 0.57–0.98 percentage points. When avoidance exiters and survivors both have passthrough of one, the correction vanishes.

Table D.2: Sensitivity of Aggregate Cohort Passthrough (2023)

Assumed passthrough		Cohort passthrough		Survivor–cohort (p.p.)	
β_{survivor}	$\bar{\beta}_a$	Low screen	High screen	Low screen	High screen
<i>Panel A: Alternative avoidance-exiter passthrough</i>					
1.000	0.478	0.970	0.949	2.96	5.13
1.000	0.567	0.975	0.957	2.46	4.26
1.000	0.900	0.994	0.990	0.57	0.98
1.000	1.000	1.000	1.000	0.00	0.00
<i>Panel B: Lower survivor passthrough</i>					
0.816	0.525	0.800	0.788	1.56	2.77
0.874	0.525	0.855	0.840	1.92	3.37
<i>Panel C: Trade-value weighting</i>					
1.000	0.525	0.977	0.960	2.28	4.01

Notes: Participation-exiter passthrough is one throughout. Panels A–B use exposure-adjusted weights; Panel C uses 2017 import-value weights. The survivor–cohort difference is $100(\beta_{\text{survivor}} - \beta_{\text{cohort}})$, in percentage points.

Panel B holds fixed the passthrough of transhippers and instead alters survivor passthrough to 0.816 and 0.874 using the continuing-relationship estimates from Alviarez et al. (2023).³⁴ Be-

33. At the margin, passthrough of 0.525 implies $|\kappa|/(\varepsilon - 1) = 1/0.525 - 1 \approx 0.905$. At a demand elasticity of five, this corresponds to a superelasticity of approximately 3.62, close to the value of four used by Gopinath and Itskhoki (2010), which implies an own-cost passthrough factor of 0.50.

34. Column (1) of Table 3 reports a duty-exclusive price coefficient of -0.126 , while column (1) of Appendix Table D.2 reports -0.184 under more stringent fixed effects. Adding one gives buyer-price passthrough of 0.874 and 0.816, respectively.

cause participation-exiter passthrough remains one, participation now offsets part of the avoidance contribution: participation exiters have higher passthrough than survivors, while avoidance exiters have lower passthrough. The net correction nevertheless remains downward under both screens, ranging from 1.56 to 3.37 percentage points across these scenarios. Finally, Panel C replaces exposure-adjusted weights with 2017 import-value weights while retaining the central passthrough calibration. The resulting survivor-cohort difference is 2.28–4.01 percentage points.

Supplemental Appendix

S.1 Supplemental Theoretical Results

S.1.1 Cross-Side Equivalence

The baseline model assigns channel-choice and price decisions to the exporter. In practice, U.S. importers and trade intermediaries also have incentives to arbitrage across direct and transshipped channels. The following lemma shows that the channel partition characterized in Proposition 7 is invariant to which side of the transaction bears the fixed costs, provided the parties bargain efficiently and can transfer surplus between them.

Lemma 2 (Cross-side equivalence). *Under D1–D5 and Assumption 1, consider an exporter with marginal cost m and a U.S. importer who resells competitively at the delivered price $(1 + \tau^r)p$. Suppose the parties bargain efficiently over channel choice, operation, and p , and can make lump-sum transfers that do not change demand or tariff payments. Normalize their outside payoffs to zero. Their optimized joint payoffs are*

$$J_d(m) = V(m, 1 + \tau^d) - F_d, \quad J_a(m) = V(m, 1 + \tau^a) - F_d - F_a.$$

The channel partition and cutoffs of Proposition 7 are unchanged: $\Delta V(m_a) = F_a$ and $V(m_p, 1 + \tau^d) = F_d$. They do not depend on how the fixed costs are allocated between the parties.

Proof. Competitive resale adds no variable margin, so joint variable surplus is $(p - m)D((1 + \tau^r)p | P)$. Transfers between the parties cancel from their combined payoff, and the total fixed cost is unchanged by its allocation. Maximizing over p therefore yields $V(m, 1 + \tau^r)$ and the joint payoffs above. Efficient bargaining with transferable surplus selects the largest of $J_d(m)$, $J_a(m)$, and zero. This is exactly the choice problem in equation (17), so Proposition 7 gives the same cutoffs and partition. The bargaining protocol determines only how the maximized surplus is divided. $\square$

Since the price and channel choices are unchanged, so are the associated margin-specific selection contributions. The lemma does not establish equivalence under bargaining frictions that change those choices.

S.1.2 Heterogeneous Fixed Costs

This appendix allows the transshipment setup cost F_a and the direct-channel operating cost F_d to vary across firms or products. The model's partition and opposing selection across exit margins

continue to hold conditional on a fixed-cost pair. Pooling these groups requires distinguishing that conditional ordering from the ordering of aggregate exiter and survivor passthrough.

Corollary 3 (Heterogeneous fixed costs). *Suppose D1–D5 hold. Let fixed costs $(F_{a,i}, F_{d,i})$ vary across firms but remain unchanged when τ^d rises, holding τ^a and $\mathbf{P}$ fixed. Assume that each fixed-cost pair satisfies Assumption 1 before and after the tariff increase. At cutoff indifference, firms remain in direct trade.*

(i) **Conditional cutoffs and partition.** *For each pair (F_a, F_d) , the unique cutoffs satisfy*

$$\Delta V(m_a(F_a)) = F_a, \quad V(m_p(F_d), 1 + \tau^d) = F_d, \quad \underline{m} < m_a(F_a) < m_p(F_d) < \bar{m}.$$

A firm transships below its avoidance cutoff, ships directly between its cutoffs, and exits above its participation cutoff. The tariff comparative statics of Proposition 7 hold conditional on (F_a, F_d) ; additionally, $\partial m_a / \partial F_a < 0$ and $\partial m_p / \partial F_d < 0$.

(ii) **Selection within fixed-cost groups.** *Restrict attention to the original cohort of direct shippers and use its baseline transaction-value weights. Within each fixed-cost group, provided the groups being compared have positive baseline value, new transshippers have weakly lower average direct-channel passthrough than survivors, while participation exiters have weakly higher average passthrough. The inequalities are strict under $\kappa < 0$.*

(iii) **Aggregate comparison.** *Let θ^a and θ^p be the shares of the original cohort's baseline value that exit through each margin, and let $\bar{\beta}_a$, $\bar{\beta}_p$, and β_{survivor} be the corresponding pooled averages, as in Section 3. When both margins are present and $\theta^a + \theta^p < 1$,*

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \theta^a (\beta_{\text{survivor}} - \bar{\beta}_a) - \theta^p (\bar{\beta}_p - \beta_{\text{survivor}}).$$

The sign of this expression determines which bound in Proposition 2 applies. The margin-specific bounds in Proposition 5 require restrictions on these pooled exiter averages; the conditional ordering in (ii) alone does not establish those restrictions.

Proof. Conditional on (F_a, F_d) , Proposition 7 applies at both tariff levels. Differentiating the cutoff equations gives $\partial m_a / \partial F_a = 1 / \Delta V_m < 0$ and $\partial m_p / \partial F_d = 1 / V_m < 0$. A tariff increase raises m_a and lowers m_p , removing the low- and high-cost tails of the group's original direct shippers. Proposition 6 then gives (ii), and the cohort accounting in equation (9) gives (iii). $\square$

The conditional results do not require independence between fixed costs and marginal cost. Conversely, independence in the population does not generally survive restriction to the original direct-trade cohort: membership requires marginal cost to lie between cutoffs that depend on the

fixed-cost pair. Mean passthrough can therefore differ across fixed-cost groups, and those groups can account for different shares of exiters and survivors. The aggregate dominance comparison must use their pooled averages.

If fixed costs are common within an end-use category, the same results apply within that category. Lower transshipment setup costs raise the avoidance cutoff, while lower direct operating costs raise the participation cutoff. Fixed-cost differences can therefore generate different exit patterns across categories without changing the conditional sorting mechanism.

S.1.3 Variable Transshipment Costs

Section 3.1 incorporates costs proportional to shipment value into the avoidance wedge. An additive per-unit cost is different. Suppose a common cost $\Delta c \geq 0$ raises marginal cost in the transshipment channel from m to $m + \Delta c$. The net profit advantage of transshipment over direct trade is

$$H(m, \Delta c) \equiv V(m + \Delta c, 1 + \tau^a) - V(m, 1 + \tau^d) - F_a.$$

Corollary 4 (Selection under variable transshipment costs). *Suppose **D1–D5** hold on the enlarged cost range $[\underline{m}, \bar{m} + \Delta c]$, and Assumption 1 holds at $\Delta c = 0$. Let $q^r(x)$ denote optimized quantity in channel r at marginal cost x . If $q^a(m + \Delta c) > q^d(m)$ for $m \in [\underline{m}, m_p]$ and $H(\underline{m}, \Delta c) > 0$, there is a unique cutoff $m_{a,\Delta} \in (\underline{m}, m_p)$ satisfying $H(m_{a,\Delta}, \Delta c) = 0$. Firms transship below $m_{a,\Delta}$, ship directly between $m_{a,\Delta}$ and m_p , and exit above m_p . The avoidance cutoff increases with τ^d and decreases with τ^a , F_a , and Δc . The participation cutoff m_p is unaffected by Δc .*

Proof. The envelope theorem gives $H_m = -q^a(m + \Delta c) + q^d(m) < 0$ on $[\underline{m}, m_p]$. The lower-end value is positive by assumption, while $H(m_p, \Delta c) \leq H(m_p, 0) < 0$ by $V_m < 0$ and Assumption 1. Continuity and strict monotonicity therefore give a unique cutoff. Below m_p , direct profit is nonnegative and the sign of H determines the preferred channel. Above m_p , both baseline channel profits are negative, and the added cost can only reduce transshipment profit.

At the cutoff, $H_{\tau^d} > 0$, $H_{\tau^a} < 0$, $H_{F_a} = -1$, and $H_{\Delta c} = -q^a(m + \Delta c) < 0$. The implicit function theorem gives the stated comparative statics; in particular,

$$\frac{\partial m_{a,\Delta}}{\partial \Delta c} = \frac{q^a(m_{a,\Delta} + \Delta c)}{H_m(m_{a,\Delta}, \Delta c)} < 0.$$

The direct-channel break-even condition is unchanged, so m_p is unchanged. □

A positive Δc moves the avoidance cutoff leftward relative to the zero-cost benchmark. For direct-tariff increases that preserve these conditions with Δc fixed, new transshippers still come from the low-cost tail of the original direct shippers, and participation exiters from the high-cost

tail. The opposing selection directions survive, but changing the size of avoidance can change which margin dominates in equation (19).

S.1.4 Heterogeneous Avoidance Wedges

Let the effective avoidance wedge τ_k^a vary across products but remain common across firms within each product. As in Section 3.1, it includes statutory hub tariffs and additional costs proportional to shipment value. The empirical hub tariff τ_k^h measures only the statutory component, so the observed tariff gap $\Delta_k = \log[(1 + \tau_k^d)/(1 + \tau_k^h)]$ does not incorporate all transshipment costs. Additive per-unit costs are treated separately in Appendix S.1.3.

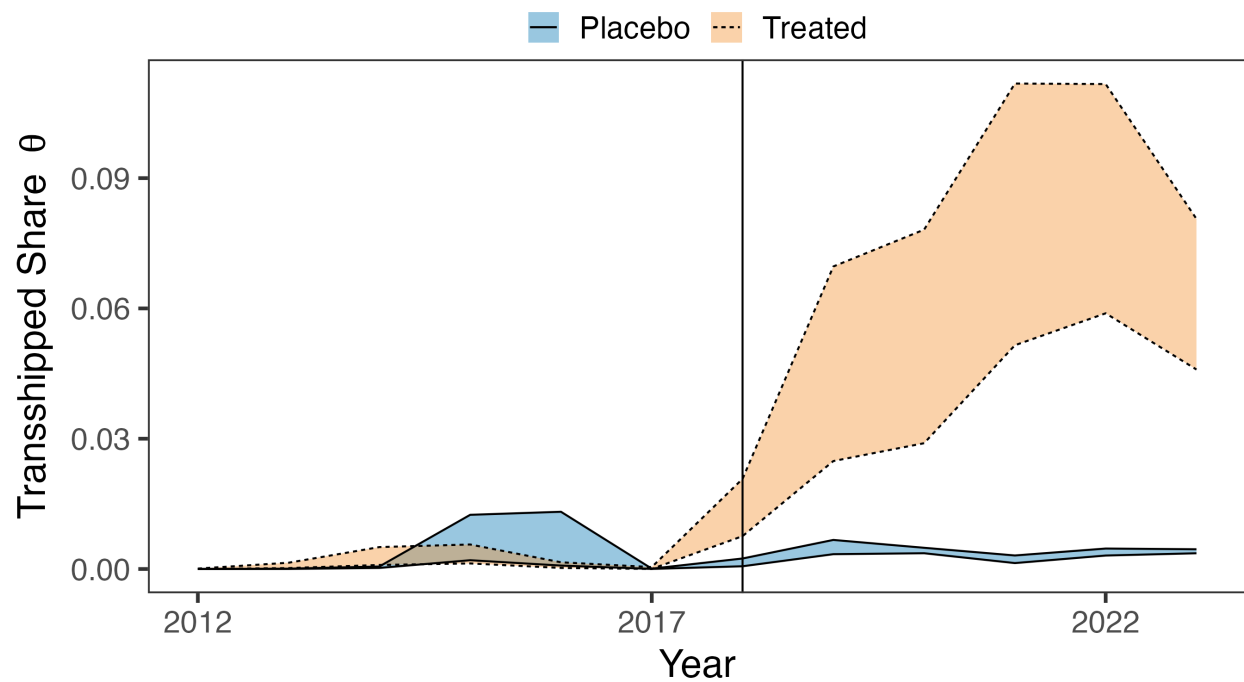
Under **D1–D5** and Assumption 1 within each product, Proposition 7 gives cutoffs satisfying $\Delta V_k(m_{a,k}) = F_a$ and $V(m_{p,k}, 1 + \tau_k^d) = F_d$. Holding the direct tariff and other primitives fixed, a higher avoidance wedge lowers $\Delta V_k(m)$ at every marginal cost and therefore lowers $m_{a,k}$, while leaving $m_{p,k}$ unchanged. A higher direct tariff instead raises $m_{a,k}$ and lowers $m_{p,k}$. These comparisons apply while the interior-cutoff conditions continue to hold.

To translate these cutoff movements into exit shares, hold fixed the original direct-trade cohort and its baseline transaction values. A lower avoidance cutoff then implies a weakly smaller θ_k^a : the share of that cohort's baseline value associated with varieties that subsequently transship, as defined in Section 4.2, rather than the fraction of all exporters below the cutoff. Thus, holding direct tariffs and other route costs fixed, higher hub tariffs predict less transshipment. The resulting exit shares enter the two-margin bounds in Proposition 5 under its stated restrictions on average exporter passthrough.

S.2 Supplemental Empirical Results

S.2.1 Measurement Checks

Figure S1: Aggregate Transshipment among Treated and Untreated HS6 Codes



Notes: Lines show the low and high transshipment shares for tariff-exposed and placebo products, weighted by their 2017 China–U.S. import values within each group. Shading spans the low and high screens. Measures are constructed as in Section B.

Table S1: Transshipment Response to Tariffs under Alternative Route-Eligibility Rules

Route-eligibility rule	Transshipment share		Any transshipment	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
Baseline	0.236*** (0.054)	0.474*** (0.112)	1.37 (0.961)	1.37 (0.961)
No tariff gate	0.236*** (0.054)	0.474*** (0.112)	1.37 (0.961)	1.37 (0.961)
Fixed 2019 eligibility	0.237*** (0.054)	0.476*** (0.112)	1.38 (0.961)	1.38 (0.961)
Observations	52,872	52,872	52,872	52,872
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

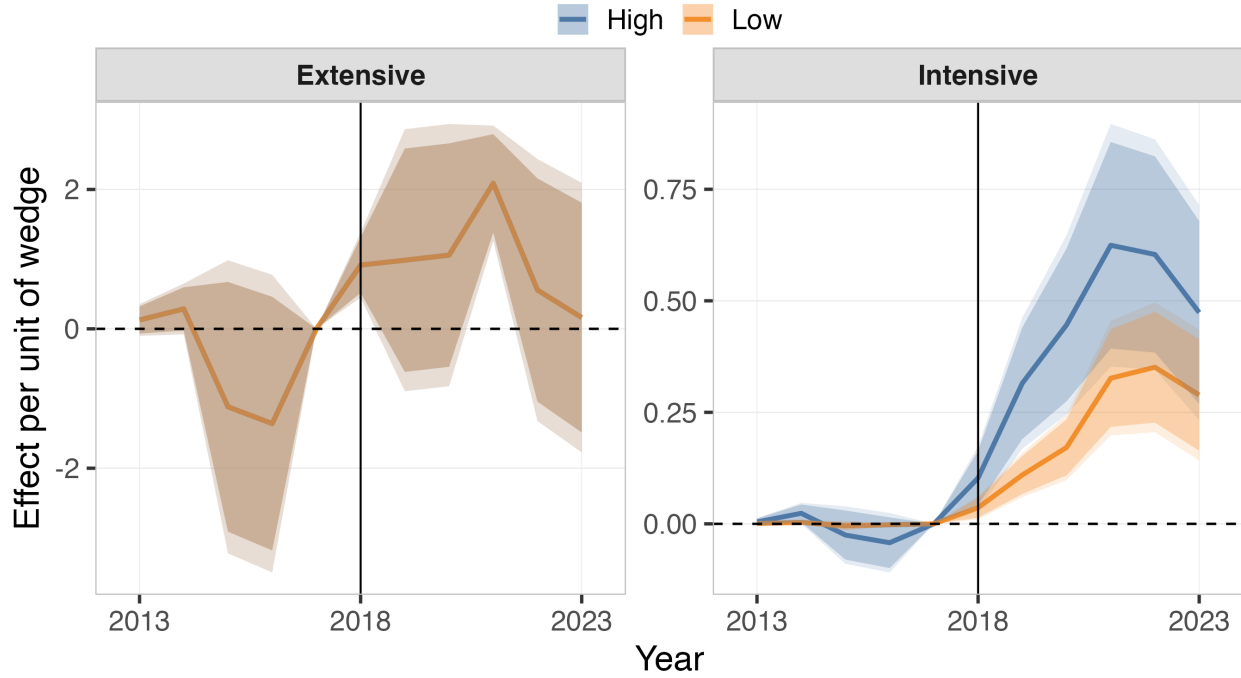
* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Entries report coefficients on $\Delta_{k,t}$; HS6-clustered standard errors are in parentheses. The baseline uses the production route-eligibility rule. The no-tariff-gate row removes only its tariff-feasibility condition, while the fixed-2019 row applies the production 2019 tariff-feasibility indicator in every year. All other construction and regression choices are held fixed. Regressions include HS6 and year $\times$ end-use fixed effects and weight by 2017 China-to-U.S. exposure times pre-period end-use share.

S.2.2 Tariff-Response Robustness

Figure S2: Event Study: Transshipment Response to Tariff Exposure

$$\theta_{k,t}^a = \alpha_k + \delta_{t \times e(k)} + \sum_{r \neq -1} \beta_r \mathbf{1}\{\text{rel}_t = r\} \times \Delta_{k,2019} + \varepsilon_{k,t}. \quad (\text{S.1})$$



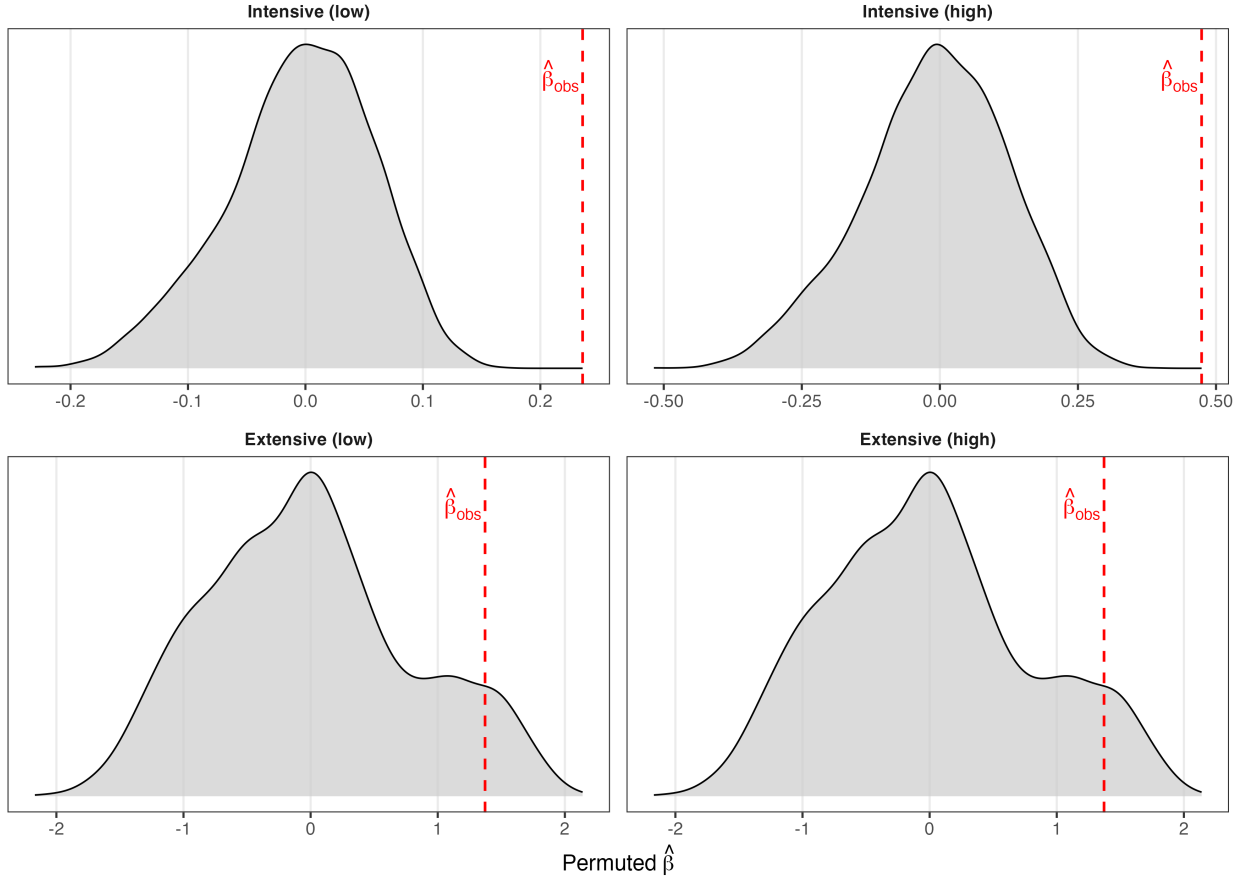
Notes: Coefficients β_r from equation (S.1). Exposure $\Delta_{k,2019}$ is the product's 2019 direct-to-hub tariff wedge. Event time rel_t is relative to 2018, with 2017 omitted and 2012–2013 pooled in the first bin. The left panel uses $\mathbf{1}(\theta_{k,t}^a > 0)$; the right uses $\theta_{k,t}^a$. Regressions include HS6 fixed effects α_k and year $\times$ end-use fixed effects $\delta_{t \times e(k)}$, with 2017 import-value weights allocated by end use. Shading shows 90% and 95% confidence intervals based on HS6-clustered standard errors. Low screen in orange; high screen in blue. Supplemental Figure S5 reports estimates using alternative exposure years.

Table S2: Permutation Tests for Pre-Period Balance

Outcome	Observed gap	Left-tail p	Right-tail p	Two-sided p
Transshipment share, low screen	-0.0001	0.281	0.719	0.563
Transshipment share, high screen	-0.0017	0.223	0.777	0.446
Any transshipment	-0.0787	0.249	0.751	0.498

Notes: Each row reports the 2012–2017 treated-minus-placebo weighted mean gap and permutation p -values. The 5,000 permutations reassign treatment labels within HS4 groups, preserving each group’s observed number of treated products. Outcomes are the low and high transshipment shares and an indicator of any detected transshipment.

Figure S3: Randomization Inference: Null Distributions of $\hat{\gamma}$ under Permuted Wedge Assignment



Notes: Each of 5,000 permutations randomly reassigns wedge trajectories across HS6 codes. Each product k inherits the full time series $\{\Delta_{k',t}\}_t$ from a donor product k' , preserving within-product wedge dynamics but randomizing which product faces which trajectory. Equation (20) is re-estimated with HS6 and year $\times$ end-use fixed effects, 2017 China $\rightarrow$ US import-share weights, and HS6 clustering. Gray densities plot the null distributions of $\hat{\gamma}$. Red dashed lines mark the observed $\hat{\gamma}$. Top panels use intensive-margin outcomes θ_{low}^a and θ_{high}^a . Bottom panels use extensive-margin outcomes $1(\theta_{\text{low}}^a > 0)$ and $1(\theta_{\text{high}}^a > 0)$.

Table S3: EU-27+UK and Canada Destination Placebo Results

	Transshipment share		Any transshipment	
	Low screen (1)	High screen (2)	Low screen (3)	High screen (4)
<i>EU-27+UK</i>				
Wedge (pooled)	0.019 (0.014)	0.027 (0.027)	-0.150 (0.210)	-0.150 (0.210)
Wedge (with post interaction)	0.003 (0.012)	-0.009 (0.022)	-0.509 (0.458)	-0.509 (0.458)
Wedge $\times$ post-2018	0.019* (0.011)	0.042* (0.022)	0.423 (0.480)	0.423 (0.480)
R ² (pooled)	0.32	0.33	0.29	0.29
R ² (with post interaction)	0.32	0.33	0.29	0.29
Observations	54,612	54,612	54,612	54,612
<i>Canada</i>				
Wedge (pooled)	0.098*** (0.036)	0.108* (0.056)	0.284 (0.233)	0.284 (0.233)
Wedge (with post interaction)	0.152* (0.080)	0.194 (0.119)	0.058 (0.410)	0.058 (0.410)
Wedge $\times$ post-2018	-0.066 (0.072)	-0.105 (0.107)	0.276 (0.468)	0.276 (0.468)
R ² (pooled)	0.29	0.30	0.33	0.33
R ² (with post interaction)	0.29	0.30	0.33	0.33
Observations	52,764	52,764	52,764	52,764
HS6 & Year $\times$ Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Outcomes measure transshipment to the indicated destination. The regressor is the U.S. direct-to-hub tariff wedge. The pooled specification includes the wedge alone; the post-interaction specification adds its interaction with an indicator for 2018 onward. All regressions include HS6 and year $\times$ end-use fixed effects and use 2017 import-value weights allocated by end use. HS6-clustered standard errors are in parentheses.

Table S4: Transshipment Response to Tariffs (Clustered by HS4 and Year)

	Transshipment share		Any transshipment	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
Aggregate	0.236*** (0.059)	0.474*** (0.116)	1.37 (1.16)	1.37 (1.16)
Capital	0.315** (0.107)	0.676*** (0.205)	1.52 (1.72)	1.52 (1.72)
Consumption	0.151*** (0.038)	0.297*** (0.075)	1.16 (1.13)	1.16 (1.13)
Intermediate	0.245*** (0.073)	0.423** (0.142)	1.48*** (0.366)	1.48*** (0.366)
R ² (aggregate)	0.41	0.46	0.42	0.42
R ² (end use)	0.41	0.46	0.42	0.42
Observations	52,872	52,872	52,872	52,872
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The regressor is the log gross direct-to-hub tariff wedge. Regressions include HS6 and year × end-use fixed effects and use 2017 import-value weights allocated by BEC end-use shares. Standard errors clustered by HS4 and year are in parentheses. Aggregate and end-use coefficients come from separate specifications.

Table S5: Transshipment Response to Tariffs with Alternative Fixed Effects

	Transshipment share		Any transshipment	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
Aggregate	0.321*** (0.077)	0.659*** (0.139)	1.33 (1.19)	1.33 (1.19)
Capital	0.332** (0.108)	0.711*** (0.206)	1.67 (1.48)	1.67 (1.48)
Consumption	0.212*** (0.058)	0.429*** (0.118)	1.06 (1.30)	1.06 (1.30)
Intermediate	0.336*** (0.073)	0.668*** (0.124)	1.13 (0.937)	1.13 (0.937)
R ² (aggregate)	0.43	0.48	0.49	0.49
R ² (end use)	0.43	0.49	0.49	0.49
Observations	52,872	52,872	52,872	52,872
HS6 & Year & Year×HS2 FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The regressor is the log gross direct-to-hub tariff wedge. Regressions include HS6, year, and HS2 × year fixed effects and use 2017 import-value weights allocated by BEC end-use shares. Standard errors clustered by HS4 and year are in parentheses. Aggregate and end-use coefficients come from separate specifications.

Table S6: Transshipment Response to Tariffs (unweighted)

	Transshipment share		Any transshipment	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
Aggregate	0.111*** (0.017)	0.181*** (0.024)	0.394*** (0.060)	0.394*** (0.060)
Capital	0.226*** (0.045)	0.382*** (0.066)	0.685*** (0.168)	0.685*** (0.168)
Consumption	0.102*** (0.026)	0.165*** (0.035)	0.407*** (0.105)	0.407*** (0.105)
Intermediate	0.089*** (0.020)	0.143*** (0.026)	0.325*** (0.057)	0.325*** (0.057)
R ² (aggregate)	0.33	0.36	0.41	0.41
R ² (end use)	0.33	0.36	0.41	0.41
Observations	52,872	52,872	52,872	52,872
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted				

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The regressor is the log gross direct-to-hub tariff wedge. Regressions include HS6 and year × end-use fixed effects and use no observation weights. HS6-clustered standard errors are in parentheses. Aggregate and end-use coefficients come from separate specifications.

Table S7: Transshipment Response to the Direct Tariff Alone

	Transshipment	
	Low screen	High screen
	(1)	(2)
Aggregate	0.222*** (0.051)	0.449*** (0.106)
Capital	0.300*** (0.094)	0.649*** (0.188)
Consumption	0.144*** (0.033)	0.285*** (0.065)
Intermediate	0.227*** (0.062)	0.402*** (0.140)
R ² (aggregate)	0.41	0.46
R ² (end use)	0.41	0.46
Observations	52,872	52,872
HS6 & Year×Use FE	✓	✓
Weighted	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The regressor is $\log(1 + \tau_{k,t}^d)$, the log gross direct tariff. Outcomes are transshipment shares. Regressions include HS6 and year × end-use fixed effects and use 2017 import-value weights allocated by BEC end-use shares. HS6-clustered standard errors are in parentheses. Aggregate and end-use coefficients come from separate specifications.

Table S8: Transshipment Response to Tariffs (2016-2019)

	Transshipment share		Any transshipment	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
Aggregate	0.136*** (0.026)	0.405*** (0.084)	1.95 (1.39)	1.95 (1.39)
Capital	0.167*** (0.030)	0.545*** (0.106)	2.21 (2.08)	2.21 (2.08)
Consumption	0.106*** (0.029)	0.311*** (0.079)	1.78 (1.47)	1.78 (1.47)
Intermediate	0.127*** (0.048)	0.302** (0.137)	1.78*** (0.435)	1.78*** (0.435)
R ² (aggregate)	0.47	0.50	0.46	0.46
R ² (end use)	0.47	0.50	0.46	0.46
Observations	17,624	17,624	17,624	17,624
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The regressor is the log gross direct-to-hub tariff wedge. Regressions include HS6 and year × end-use fixed effects and use 2017 import-value weights allocated by BEC end-use shares. HS6-clustered standard errors are in parentheses. Aggregate and end-use coefficients come from separate specifications.

Table S9: Slope-adjusted difference-in-differences

	Transshipment share		Any transshipment	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.239*** (0.054)	0.505*** (0.119)	2.41 (1.62)	2.41 (1.62)
R ²	0.41	0.46	0.48	0.48
Observations	52,872	52,872	52,872	52,872
Weighted	✓	✓	✓	✓
HS6 & Year×Use FE	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each outcome is residualized by its HS6-specific linear trend estimated over 2012–2017 before estimation. The baseline specification is then re-estimated with HS6 and year×end-use fixed effects, 2017 China→US import-share weights multiplied by pre-period end-use share, and HS6-clustered standard errors.

Table S10: Long-differenced Specification

	Transshipment share		Any transshipment		New transshipment	
	Low screen	High screen	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{k,Post} - \Delta_{k,Pre}$	0.216*** (0.047)	0.386*** (0.092)	1.60** (0.621)	1.60** (0.621)	1.30** (0.571)	1.30** (0.571)
R ²	0.13	0.16	0.13	0.13	0.09	0.09
Observations	4,081	4,081	4,081	4,081	3,791	3,791
Weighted	✓	✓	✓	✓	✓	✓
Use FE	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: For each HS6, pre means use 2012–2017 and post means use 2018–2023. Columns (1)–(4) regress long differences in the outcome on the long difference in the effective wedge with end-use fixed effects. Columns (5)–(6) restrict to HS6 codes with no pre-period activation. All regressions are weighted by 2017 import shares and cluster by HS6.

Table S11: Baseline Specification with HS4 Trends

	Transshipment share		Any transshipment	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.153*** (0.042)	0.401*** (0.101)	1.55 (1.11)	1.55 (1.11)
R ²	0.58	0.63	0.53	0.53
Observations	52,872	52,872	52,872	52,872
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓
HS4 Trend	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Specifications add linear trends for HS4 codes. Each HS6–year observation is fractionally allocated across end uses according to its UN-BEC shares. The stacked HS6–year–end-use panel includes HS6 and year × end-use fixed effects, weights by pre-period end-use share times the HS6 code’s China-to-U.S. import share, and HS6-clustered standard errors.

Table S12: The Entropy-Weighted Effect of Tariffs on Transshipment

	Transshipment share		Any transshipment	
	Low screen	High screen	Low screen	High screen
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.236*** (0.054)	0.474*** (0.112)	1.37 (0.961)	1.37 (0.961)
R ²	0.41	0.46	0.42	0.42
Observations	52,872	52,872	52,872	52,872
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Entropy balancing matches the pre-2018 mean high-screen transshipment share between products in the top quartile of the pre-2018 mean tariff wedge and the remaining products. The balancing variable is winsorized at the 1st and 99th percentiles. Regressions use 2017 exposure × entropy-balance weights × pre-period end-use share, HS6 and year×end-use fixed effects, and HS6-clustered standard errors. Coefficients are per unit of the wedge.

Figure S4: Leave-one-country-out transshipment responses

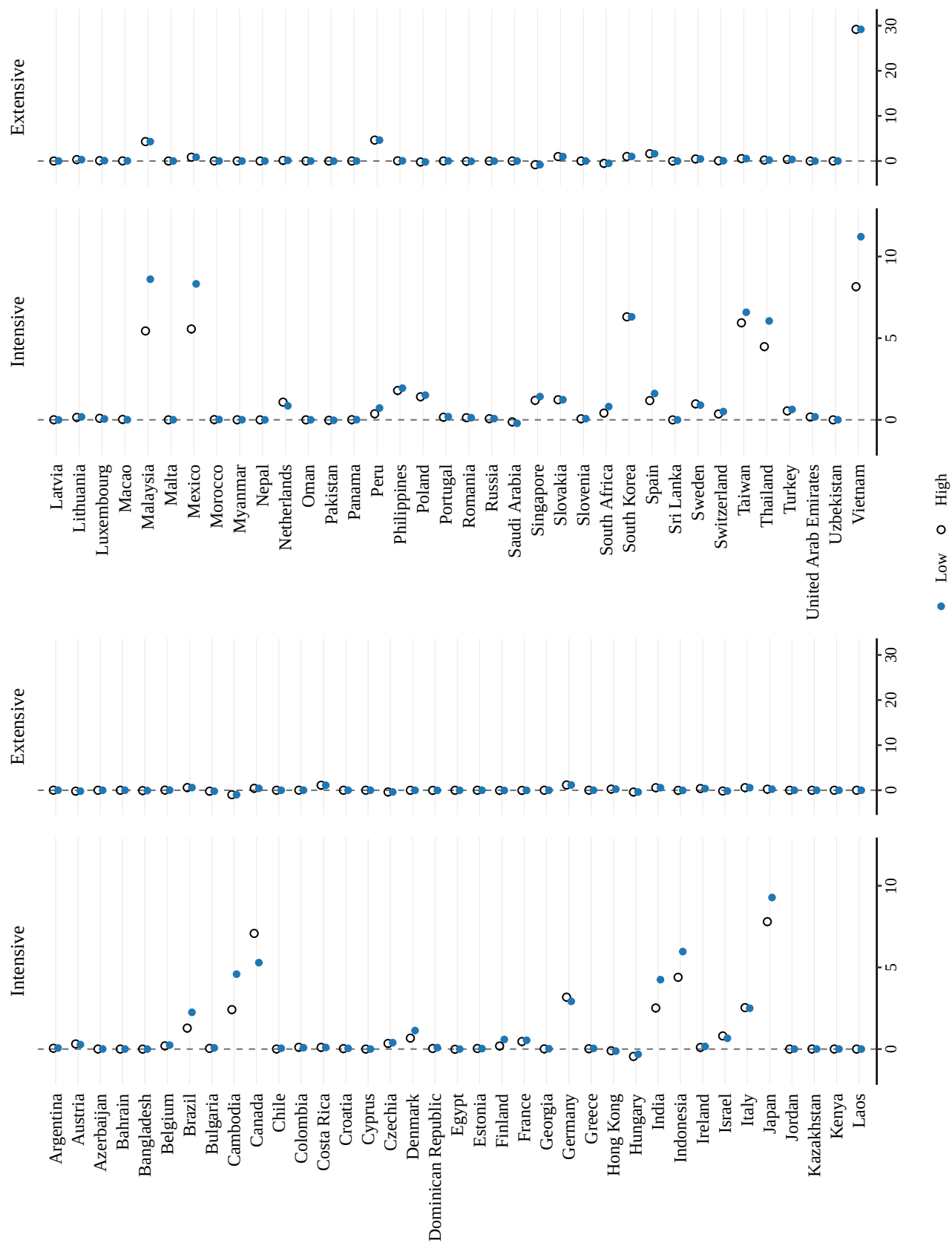
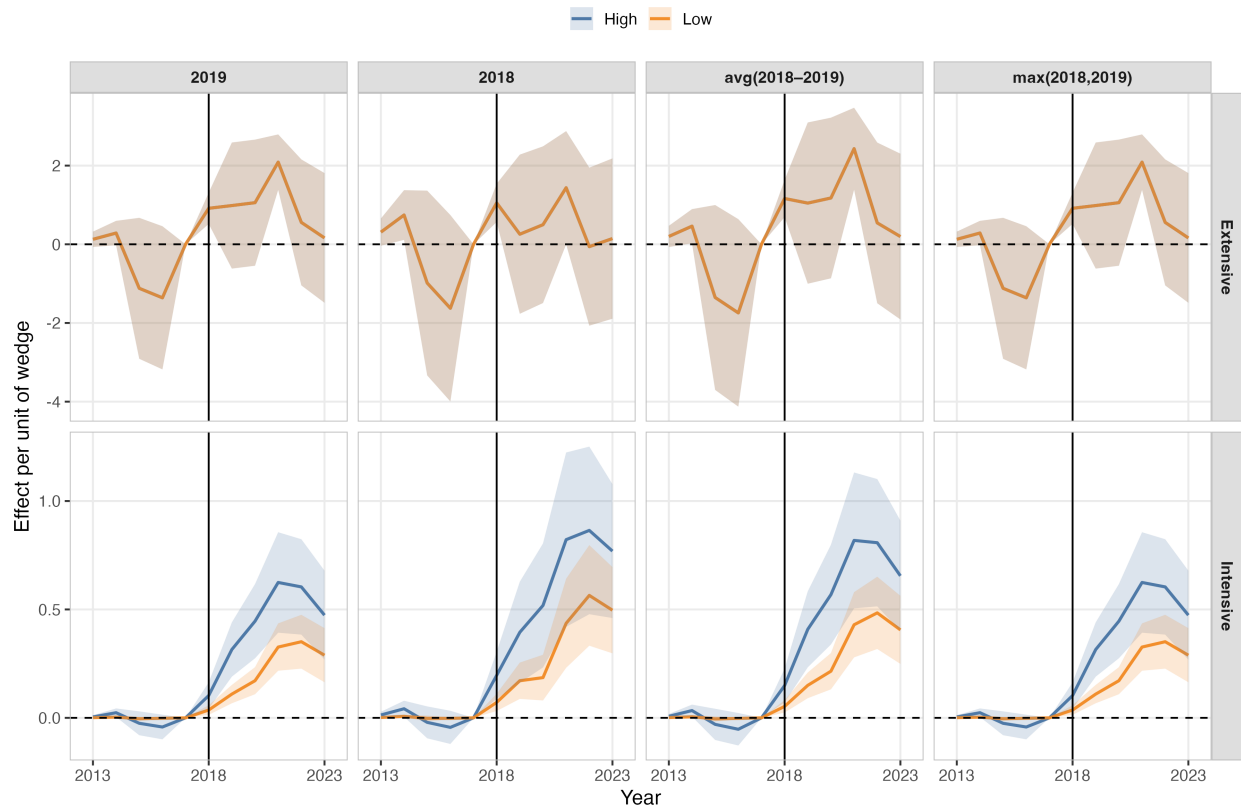


Figure S5: Transshipment Responses under Alternative Exposure Anchors



Notes: Event-study estimates under four exposure anchors: the 2019 effective wedge, the 2018 wedge, the 2018–2019 average, and the maximum of the 2018 and 2019 wedges. Panels report extensive- and intensive-margin outcomes with the same fixed effects, weights, and clustering as equation (S.1). Shading shows 90% confidence intervals.

S.2.3 Mechanism and Supporting Price Results

Table S13: Heterogeneous Fixed Costs Along the Intensive Margin

	Transshipment share					
	Low screen			High screen		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{k,t}$	0.233*** (0.052)	0.225*** (0.052)	0.190*** (0.042)	0.467*** (0.108)	0.447*** (0.108)	0.370*** (0.088)
$\Delta_{k,t} \times \text{Pre-2017 hub intensity}$	0.064 (0.047)			0.133 (0.092)		
$\Delta_{k,t} \times \text{Bulkiness}$		-0.064* (0.037)			-0.150* (0.083)	
$\Delta_{k,t} \times \text{Electronics}$			0.059** (0.029)			0.133** (0.060)
R ²	0.41	0.41	0.42	0.46	0.46	0.47
Observations	52,872	52,872	52,872	52,872	52,872	52,872
Weighted	✓	✓	✓	✓	✓	✓
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each column regresses $\theta_{k,t}^a$ on the effective wedge and its interaction with one pre-2017 product characteristic. Hub intensity is the pre-2017 hub share of China's exports of product k ; bulkiness is kilograms per U.S. dollar; electronics is an indicator for HS2 codes 84, 85, or 90. All three characteristics are standardized before interaction. All specifications include HS6 and year×use fixed effects, weight by 2017 China→US exposure times pre-period end-use share, and cluster by HS6.

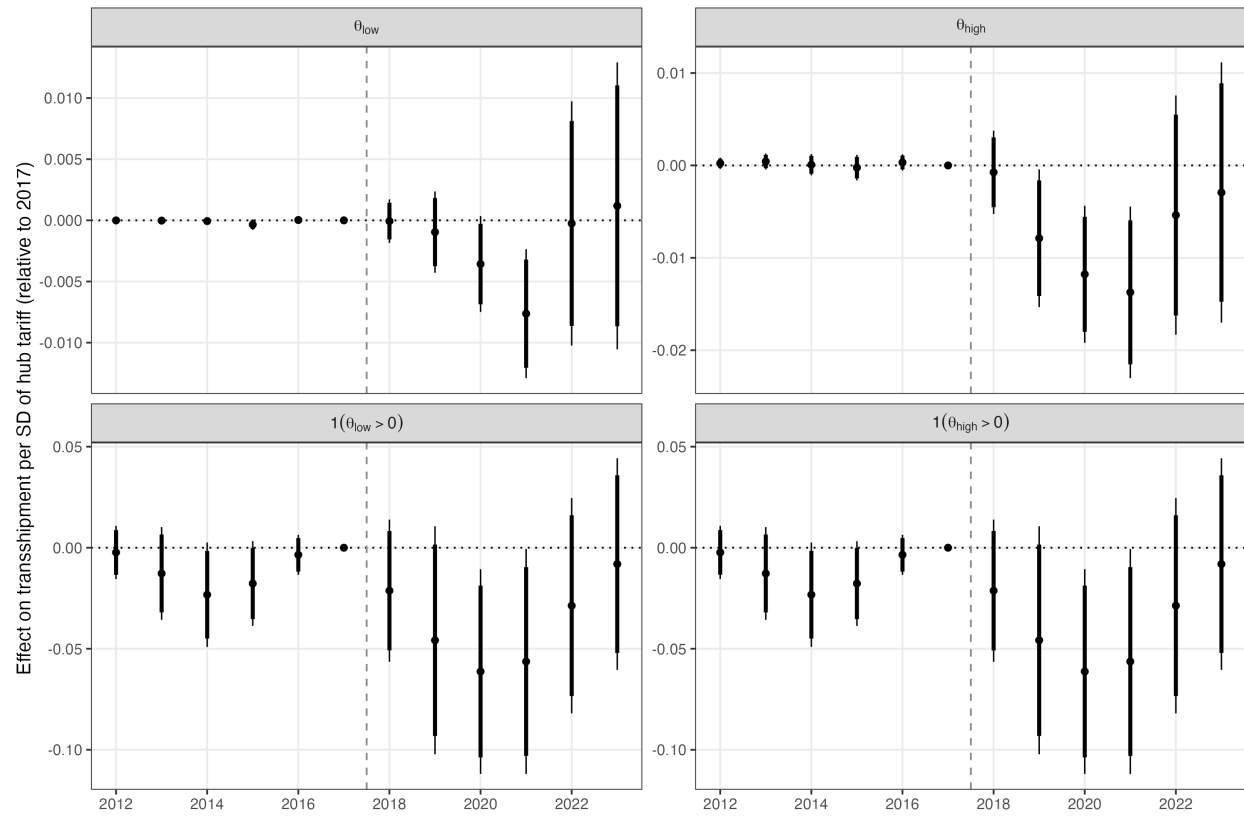
Table S14: Heterogeneous Fixed Costs Along the Extensive Margin

	Any transshipment					
	Low screen			High screen		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{k,t}$	1.38 (0.960)	1.35 (0.945)	1.33 (0.938)	1.38 (0.960)	1.35 (0.945)	1.33 (0.938)
$\Delta_{k,t} \times \text{Pre-2017 hub intensity}$	-0.096 (0.182)			-0.096 (0.182)		
$\Delta_{k,t} \times \text{Bulkiness}$		-0.100 (0.172)			-0.100 (0.172)	
$\Delta_{k,t} \times \text{Electronics}$			0.054 (0.154)			0.054 (0.154)
R ²	0.42	0.42	0.42	0.42	0.42	0.42
Observations	52,872	52,872	52,872	52,872	52,872	52,872
Weighted	✓	✓	✓	✓	✓	✓
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each column regresses the indicator for positive transshipment on the effective wedge and its interaction with one pre-2017 product characteristic. Hub intensity is the pre-2017 hub share of China's exports of product k ; bulkiness is kilograms per U.S. dollar; electronics is an indicator for HS2 codes 84, 85, or 90. All three characteristics are standardized before interaction. All specifications include HS6 and year×use fixed effects, weight by 2017 China→US exposure times pre-period end-use share, and cluster by HS6.

Figure S6: Hub Tariffs and Transshipment over Time



Notes: The sample is the tariff-exposed cohort. Points report year interactions with the standardized 2018 hub-tariff index, relative to 2017. Regressions include HS6 and year-by-end-use fixed effects and use 2017 import-value weights. Thin and thick bars show 95 and 90 percent confidence intervals based on HS6-clustered standard errors.

Table S15: Hub Tariffs and Measured Price Responses

	Census Monthly		Census Fixed basket		BACI Annual	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(1 + \tau^d)$	0.631*** (0.176)	2.405** (0.951)	0.773** (0.331)	2.586*** (0.554)	0.410 (0.335)	-0.397 (0.496)
$\Delta \log(1 + \tau^d) \times H_k$	0.106 (0.203)	-0.009 (0.132)	0.084 (0.260)	-0.063 (0.217)	0.830*** (0.247)	0.740*** (0.244)
Additional controls	No	Yes	No	Yes	No	Yes
Observations	117,264	117,264	6,045	6,045	11,121	11,121
HS6 products	2,015	2,015	2,015	2,015	3,707	3,707

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Outcomes are log duty-inclusive prices or their changes from 2017. Monthly Census observations compare each month in 2018–2019 with the same month of 2017 and include harmonized HTS10-family and calendar-time fixed effects. Annual specifications cover 2017–2019 and include HS6 and year fixed effects. H_k is the 2018 hub-tariff index standardized across the tariff-exposed cohort. Additional controls interact the tariff change with pre-tariff size, market share, routing, capacity, bulkiness, concentration, and end-use characteristics. All columns use 2017 value weights. HS6-clustered standard errors are in parentheses.

Table S16: Coverage of the Unit-value Decomposition

Quantity requirement	HS6 products	Trade coverage (%)	Exit coverage (%)			
			Transshipment		Participation	
			Low	High	Low	High
Kilograms, first field	1,540	9.6	19.5	15.4	46.7	47.3
Kilograms, either field	2,066	20.3	22.3	17.9	49.5	50.2
Common physical unit (adopted)	3,110	57.6	47.3	40.8	90.9	90.8

Notes: The decomposition uses full-year 2017 and 2019 Census values and quantities, with HTS10 categories harmonized across code changes. Census records can report quantities in two fields. The first two rows require kilograms in the first field or either field; the adopted sample permits any common physical unit from either field. Every included HS6 product must have quantities covering all recorded trade in both years in that unit. Trade coverage is the included products' share of the tariff-exposed cohort's total 2017 direct China–U.S. trade value. Exit coverage is their share of the cohort's baseline value associated with each 2019 exit margin, calculated using 2017 trade values multiplied by measured 2019 exit shares. Low and high denote the corresponding transshipment screens and participation exclusions.

Table S17: Tariffs, Exit, and Components of Unit-value Changes

	Total change		Within-category		Composition	
	Low	High	Low	High	Low	High
	screen	screen	screen	screen	screen	screen
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(1 + \tau^d)$	-0.140	-0.184	-0.082	-0.111	-0.058	-0.073
	(0.197)	(0.200)	(0.194)	(0.199)	(0.122)	(0.127)
$\Delta \log(1 + \tau^d) \times \theta_k^a$	0.098	0.524	0.889	0.713*	-0.791	-0.188
	(1.108)	(0.472)	(0.597)	(0.385)	(0.818)	(0.360)
$\Delta \log(1 + \tau^d) \times \theta_k^p$	-0.385	-0.369	-0.168	-0.143	-0.218	-0.226
	(1.126)	(1.143)	(1.208)	(1.225)	(0.266)	(0.267)
HS6 products	3,110	3,110	3,110	3,110	3,110	3,110

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each regression uses one observation per HS6 product in the adopted common-unit sample. Outcomes compare full-year 2017 and 2019 customs unit values, excluding tariffs. The within-category component measures unit-value changes among continuing harmonized HTS10 categories, holding quantity shares at their two-year averages. Composition combines changes in those shares with category entry and exit. The level components are rescaled proportionately to sum to the total log change. $\Delta \log(1 + \tau^d)$ is the change in the log gross U.S. tariff on imports from China between 2017 and 2019; θ_k^a and θ_k^p are measured 2019 transshipment and participation shares of baseline trade value, expressed from zero to one. Low and high use the corresponding screens. Each regression includes the three displayed regressors and a constant, with no additional controls or fixed effects. All columns use 2017 import-value weights. HS6-clustered standard errors are in parentheses.